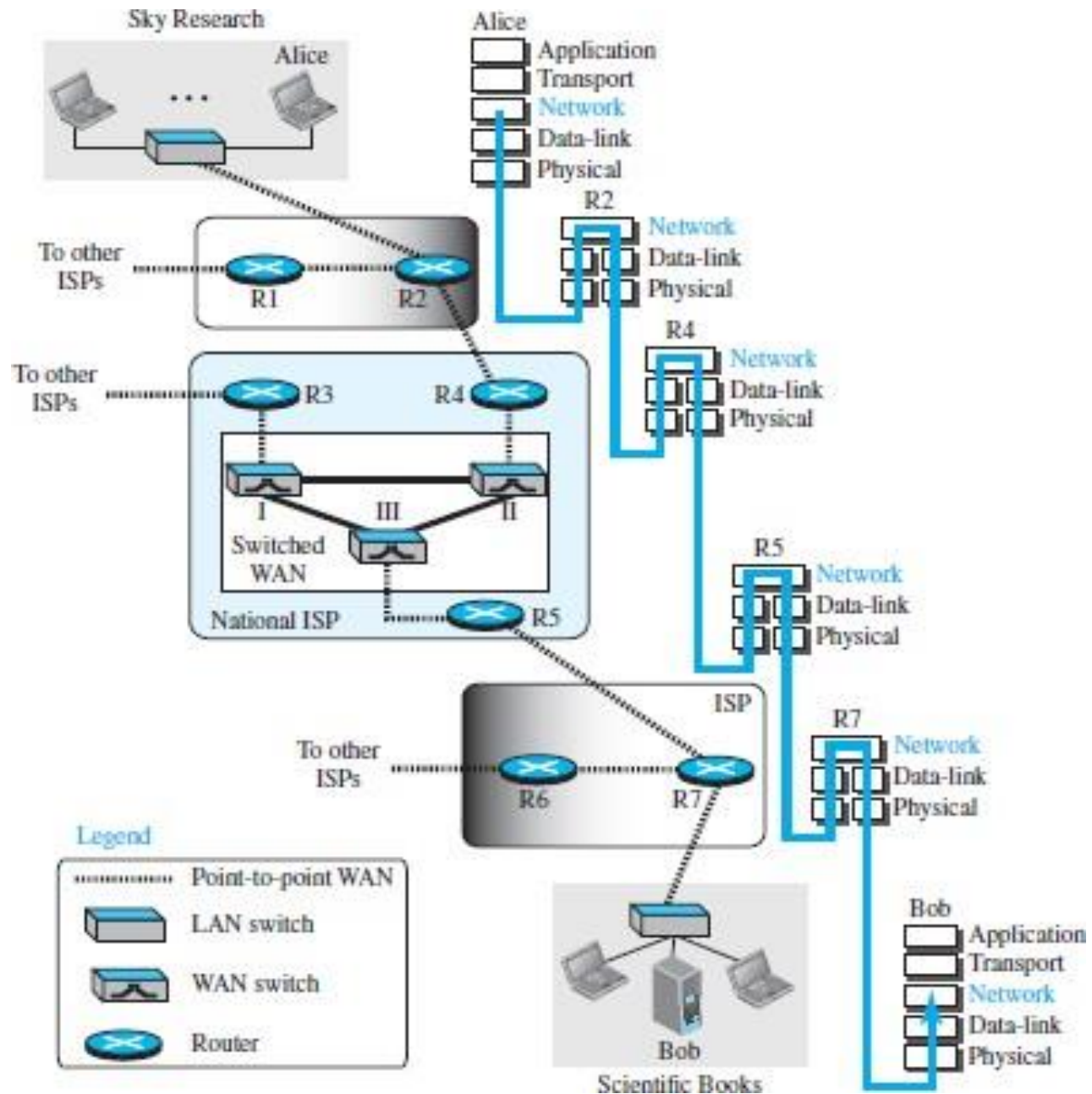


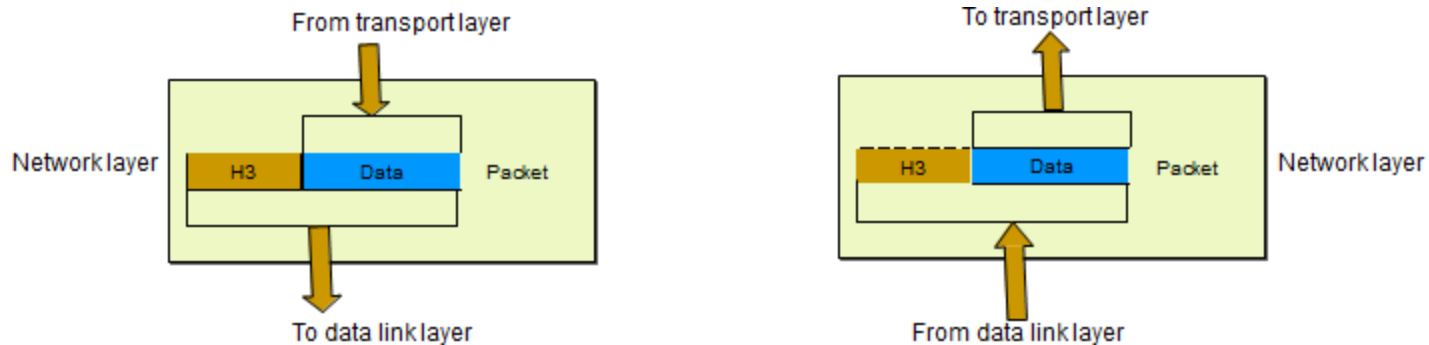
Module-2

NETWORK LAYER

Communication at the network layer



Network Layer



- Packetizing
- Routing and forwarding -Path determination
- Logical addressing
- IP is a connectionless protocol that provides no flow control, no error control, and no congestion control services.
- Protocols: IP, IGMP, ICMP etc.

Network Layer Services

- **Packetizing:** Carry a payload from the source to the destination without changing it or using it.
 - EX:** Postal office- delivery of packages from a sender to a receiver without changing or using the contents.
- The routers in the path are not allowed
 - To decapsulate the packets, to change source and destination addresses either.
 - They just inspect the addresses for the purpose of forwarding the packet to the next network on the path.

•Routing and Forwarding

Routing:

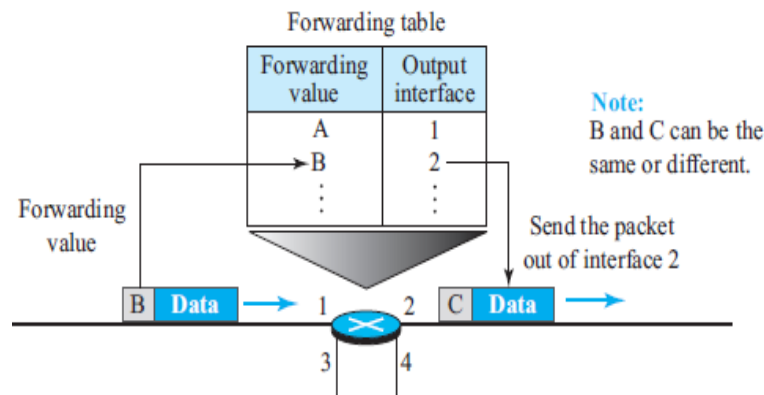
- The network layer is responsible for finding the best one among the possible routes from source to the destination.
- Done by running some routing protocols- coordinate the knowledge about the neighborhood

Network Layer Services

•Routing and Forwarding

Forwarding

- *Forwarding* can be defined as the action applied by each router when a packet arrives at one of its interfaces.
- Forwarding table - routing table.
- To make this decision, the router uses a piece of information in the packet header, which can be the destination address or a label, to find the corresponding output interface number in the forwarding table.



Other Services

- **Error Control :**
 - Added a checksum field to the datagram to control any corruption in the header
 - Error detection and Correction.
- **Flow Control :**
 - prevent a transmitter from overrunning a receiver's buffer.
- **Congestion Control : (Too many packets in network)**
 - Congestion may occur if the number of datagrams sent by source computers is beyond the capacity of the network or routers
 - System Collapses and No datagrams.
- **Quality of Service (QoS):**
 - Multimedia communication- Real time Communication.
- **Security :**
 - No security Provision
 - virtual level called IPsec (7th sem and Chapter-32)

Other Services

- **Error Control :**

Error control is ignored at network layer - the packet in the network layer may be fragmented at each router, which makes error checking at this layer inefficient

- However, designers have added a checksum field to the datagram to control any corruption in the header, but not in the whole datagram
- The Internet uses an auxiliary protocol, ICMP, that provides some kind of error control if the datagram is discarded or has some unknown information in the header

- **Flow Control :**

- prevent a transmitter from overrunning a receiver's buffer.
- The network layer in the Internet, however, does not directly provide any flow control.

- **Congestion Control : (Too many packets in network)**

- Congestion may occur if the number of datagrams sent by source computers is beyond the capacity of the network or routers
- routers may drop some of the datagrams. If it continues system Collapses and No datagrams are delivered.

- **Quality of Service (QoS):**

- Multimedia communication- Real time Communication. These provisions are mostly implemented in the upper layer

- **Security :**

- No security Provision
- virtual level called IPsec

NETWORK-LAYER PERFORMANCE

- The performance of a network can be measured in terms of:
 - **Delay**
 - **Throughput**
 - **packet loss**
- **Delay**
 - Transmission Delay
 - Propagation Delay
 - Processing Delay
 - Queuing Delay

Delay

Transmission Delay: Time it takes to place the complete data packet on the transmission medium

$$\text{Delay}_{\text{tr}} = \frac{\text{(Packet length or message size)}}{\text{(Transmission rate or bandwidth)}}$$



Fast Ethernet LAN: $(10,000)/(100,000,000)$ or 100 microseconds for all bits of the packet to be put on the line

Propagation Delay: Time it takes for a bit to go from device A to device B

$$\text{Delay}_{\text{pg}} = \frac{\text{Distance}}{\text{Propagation Speed of the media}}$$

Delay

- **Processing Delay:** the time required for a router or a destination host to receive a packet from its input port, remove the header, perform an error detection procedure, and deliver the packet to the output port

Delay_{pr} = Time required to process a packet in a router or a destination host



Queuing Delay: The queuing delay for a packet in a router is measured as the time a packet waits in the input queue and output queue of a router.

Delay_{qu} = The time a packet waits in input and output queues in a router

Total delay = (Delay_{tr} + Delay_{pg} + Delay_{pr}) + (Delay_{qu})

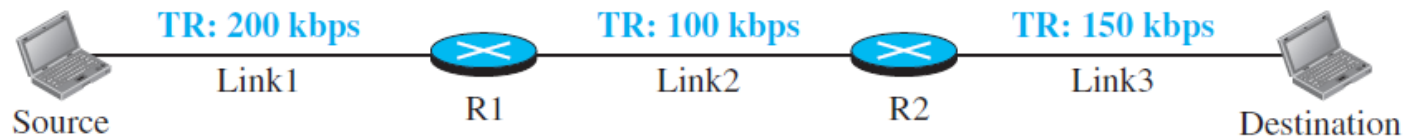
input queue: to store packets waiting to be processed

Output queue: to store packets waiting to be transmitted.

Throughput

- Bandwidth: Maximum amount of data that can be transmitted per second.
- Throughput: Actual amount of data that passes through the medium. How fast we can actually send the data through a network.

Figure 18.10 *Throughput in a path with three links in a series*

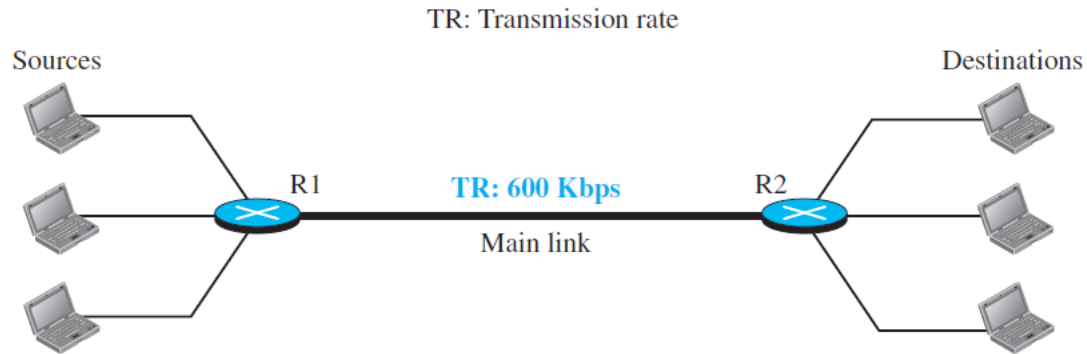


a. A path through three links

Throughput = minimum {TR1, TR2, . . . TRn}.

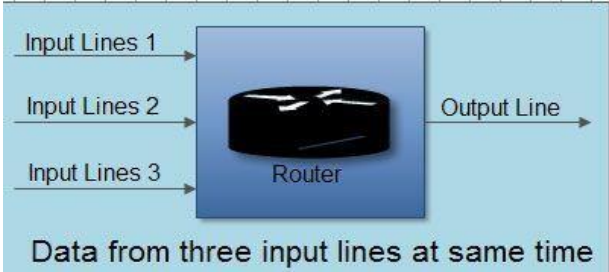
Throughput

Figure 18.12 *Effect of throughput in shared links*

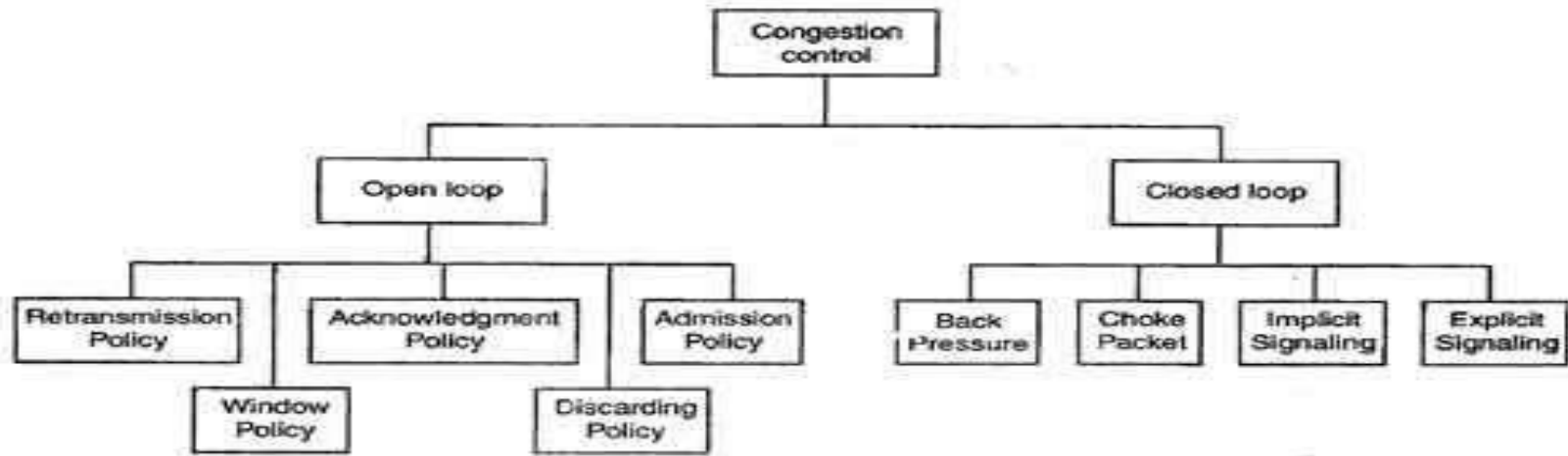


The transmission rate of the main link in the calculation of the throughput is only 200 kbps because the link is shared between three paths.

Congestion Control Algorithms

- In the network layer, when the number of packets sent to the network is greater than the number of packet the network can handle (capacity of network), a problem occurs that is known as congestion.
 - Load of the network $>$ capacity of the network
 - Fast sender and slower receiver
 - Traffic overloading
- 
- The diagram shows a central router, represented by a black cylinder with a blue square background. Three horizontal arrows labeled 'Input Lines 1', 'Input Lines 2', and 'Input Lines 3' point towards the router from the left. A single horizontal arrow labeled 'Output Line' points away from the router to the right. Below the router, the text 'Data from three input lines at same time' is written.
- Too many packets present in (a part of) the network causes packet delay and loss that degrades performance. This situation is called **congestion**.
 - **Congestion control** refers to the mechanisms and techniques that control the congestion and keep the load below the capacity.

Congestion Control Algorithms



Types of Congestion Control Methods

Open Loop Congestion Control (prevention) :

- In open-loop congestion control, policies are applied to **prevent** congestion before it happens.
- In these mechanisms, congestion control is handled by either the source or the destination.

Closed-Loop Congestion Control (removal) :

- Closed-loop congestion control mechanisms try to alleviate congestion after it happens.

Congestion Control - Open Loop Congestion Control

Retransmission Policy

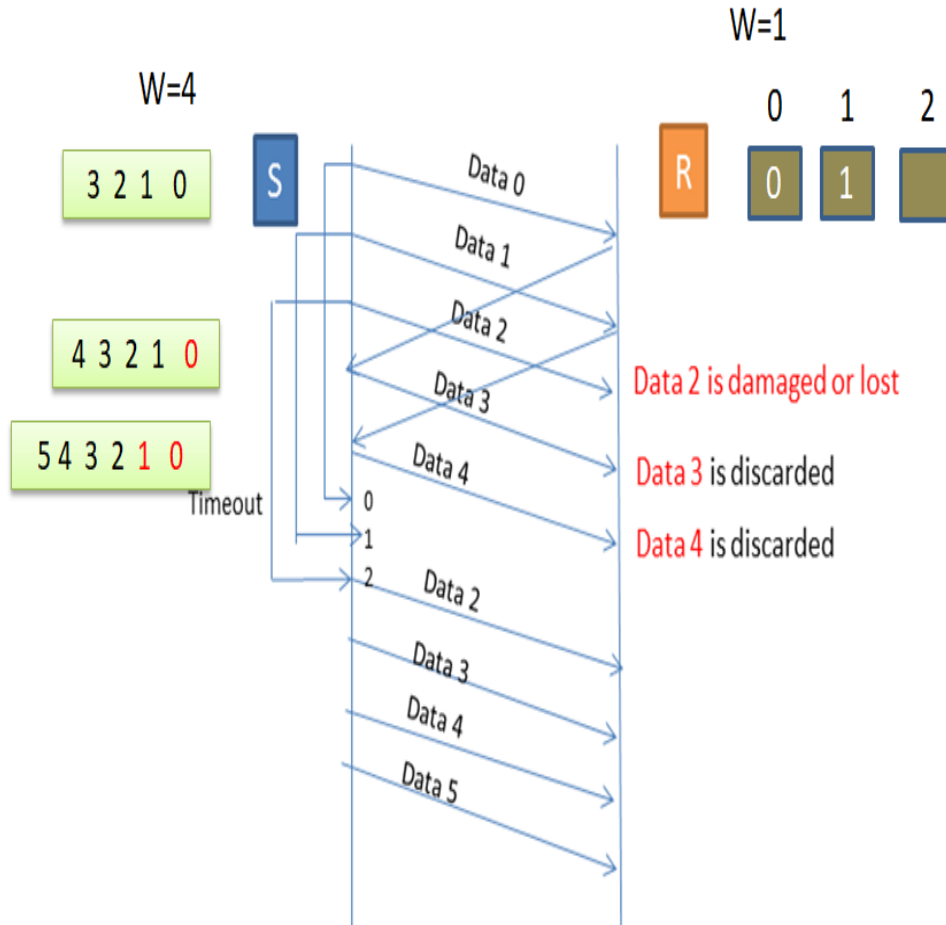
- The sender retransmits a packet, if it feels that the packet it has sent is lost or corrupted. However retransmission in general may increase the congestion in the network. But we need to implement good retransmission policy to prevent congestion.
- The retransmission policy and the retransmission timers must be designed to optimize efficiency and at the same time prevent congestion.

Window Policy

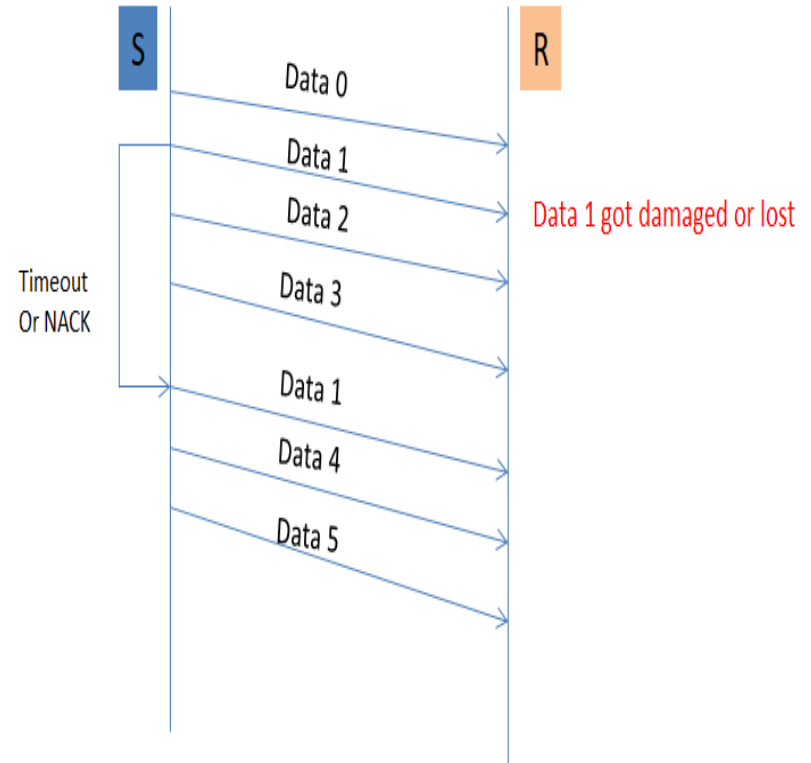
- The type of window at the sender may also affect congestion. The Selective Repeat window is better than the Go-Back-N window for congestion control.

Window Policy

Go-Back-N



Selective-Repeat



Congestion Control - Open Loop Congestion Control

Acknowledgment Policy

- The acknowledgment policy imposed by the receiver may also affect congestion.
- If the receiver does not acknowledge every packet it receives, it may slow down the sender and help prevent congestion.
- A receiver may decide to acknowledge only N packets at a time. Sending fewer acknowledgments means imposing less load on the network.

Discarding Policy

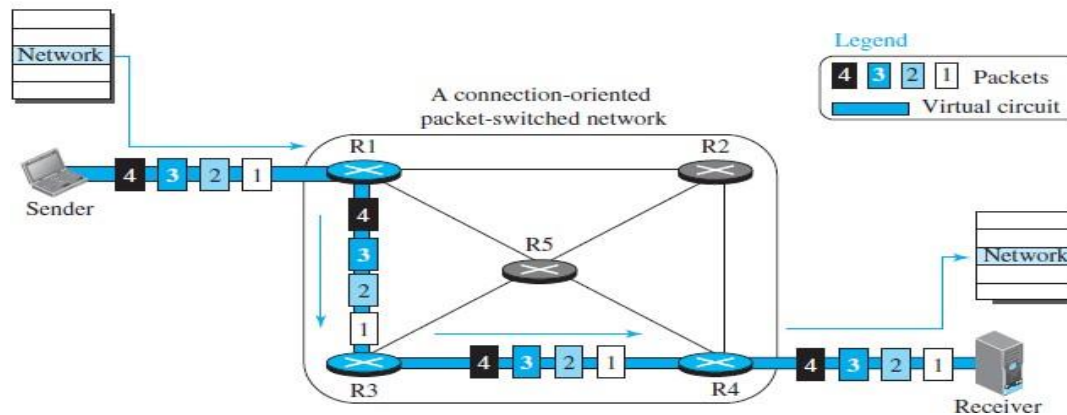
- A good discarding policy by the routers may prevent congestion and at the same time may not harm the integrity of the transmission
- A router may discard less sensitive packets when congestion is likely to happen.
- In audio transmission, if the policy is to discard less sensitive packets when congestion is likely to happen, the quality of sound is still preserved and congestion is prevented or alleviated.

Congestion Control - Open Loop Congestion Control

Admission Policy

- An admission policy, which is a quality-of-service mechanism can also prevent congestion in **virtual- circuit networks**
- Switches in a flow **first check the resource requirement** of a flow before admitting it to the network.
- A router can deny establishing a virtual-circuit connection if there is congestion in the network or if there is a possibility of future congestion.

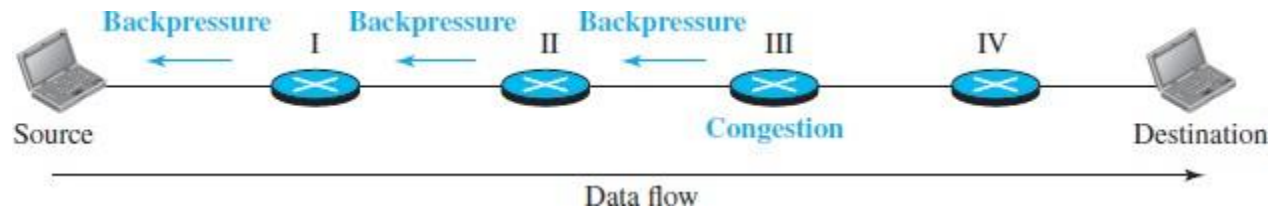
Figure 18.5 A virtual-circuit packet-switched network



Congestion Control- Closed-Loop Congestion Control

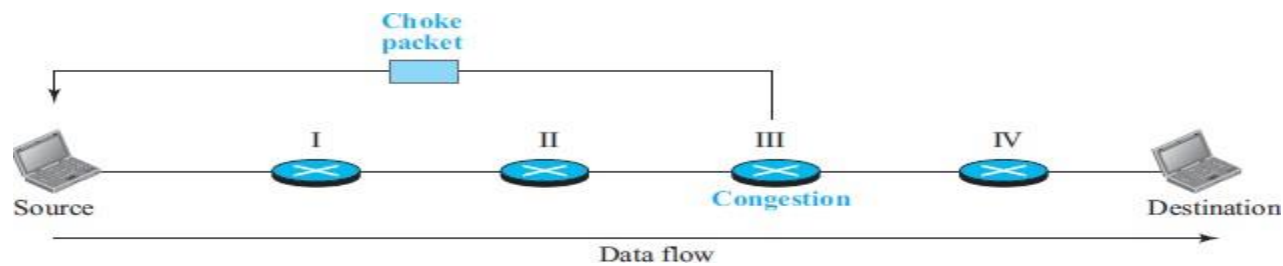
Backpressure

- Backpressure is a node-to-node congestion control that starts with a node and propagates, in the opposite direction of data flow, to the source.
- The backpressure technique can be applied only to virtual circuit networks, in which each node knows the upstream node from which a flow of data is coming.



Choke Packet

- A **choke packet** is a packet sent by a node to the source to inform it of congestion.
- The warning message goes directly to the source station; the intermediate routers do not take any action



Congestion Control- Closed-Loop Congestion Control

Implicit Signaling

- In implicit signaling, there is no communication between the congested node or nodes and the source.
- The **source guesses** that there is congestion somewhere in the network from other symptoms.
 - when a source sends several packets and there is no acknowledgment for a while.
 - The delay in receiving an acknowledgment

Explicit Signaling

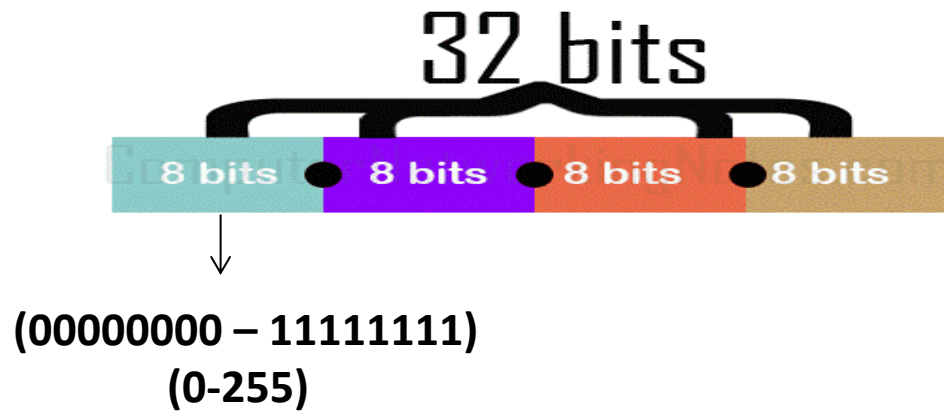
- The node that experiences congestion can **explicitly send a signal** to the source or destination.
- In the choke-packet method, a separate packet is used for this purpose; in the explicit-signaling method, **the signal is included in the packets that carry data.**
- Explicit signaling can occur in either the forward or the backward direction

IP Address (Internet Protocol Address)

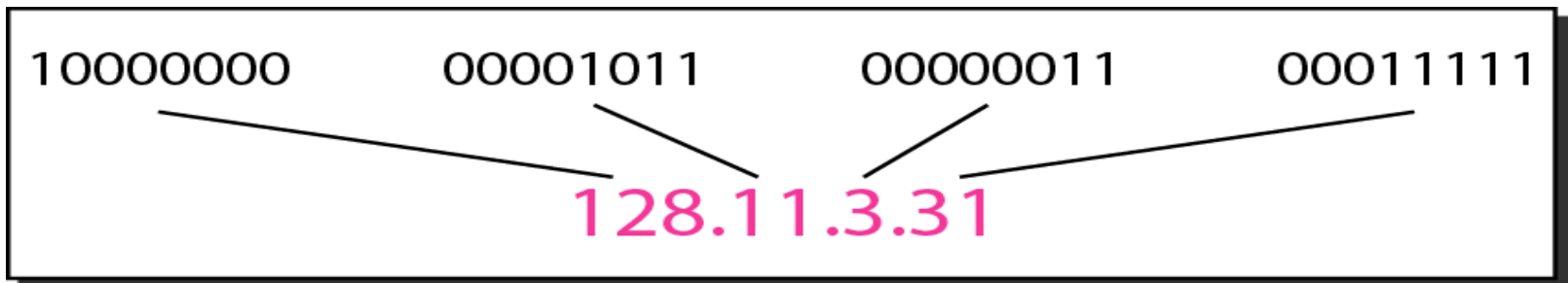
- It is an identifying number that is associated with a specific computer or computer network
- Address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.
- Two Types
 - IPV4 (32 bits)
 - IPV6 (128 bits)
- ISP provides IP address to a computer/smart phone
- Commands : ipconfig, ifconfig, website(<https://whatismyipaddress.com/>)

IPv4 address

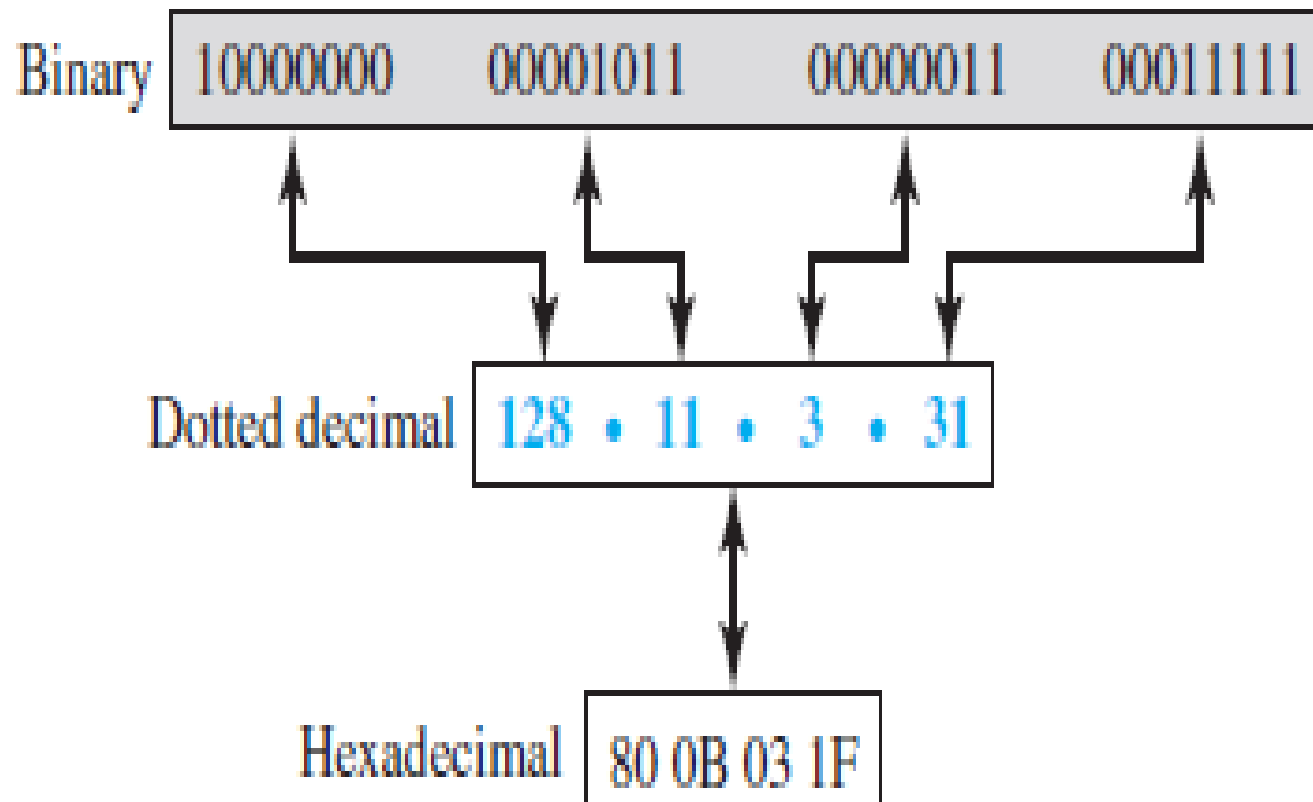
The address space of IPv4 is
 2^{32} or 4,294,967,296. (more than 4 billion)



Example



Three different notations in IPv4 addressing



IPv4 Notations

- **Binary notation:**

- Address is displayed as 32 bits.
- IPv4 address referred to as **32-bit address or 4-byte address**
- Example: **01110101 10010101 00011101 00000010**

- **Dotted-decimal notation:**

- More compact and easier to read
- Written in decimal form with a decimal point(dot) separating the bytes.
- Example: **117.149.29.2**
- Each decimal value range from **0 to 255**

- **Hexadecimal notation:**

- Easy for computer to process.
- Example: 75 95 1D 2



Example 19.1

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

We replace each group of 8 bits with its equivalent decimal number and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255



Example 19.2

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

Solution

We replace each decimal number with its binary equivalent.



Example 19.2

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

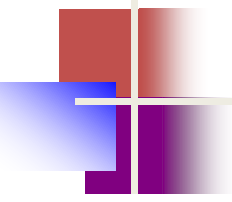
b. 221.34.7.82

Solution

We replace each decimal number with its binary equivalent.

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010

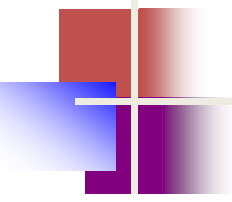


Example 19.3

Find the error, if any, in the following IPv4 addresses.

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

Solution



Example 19.3

Find the error, if any, in the following IPv4 addresses.

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

Solution

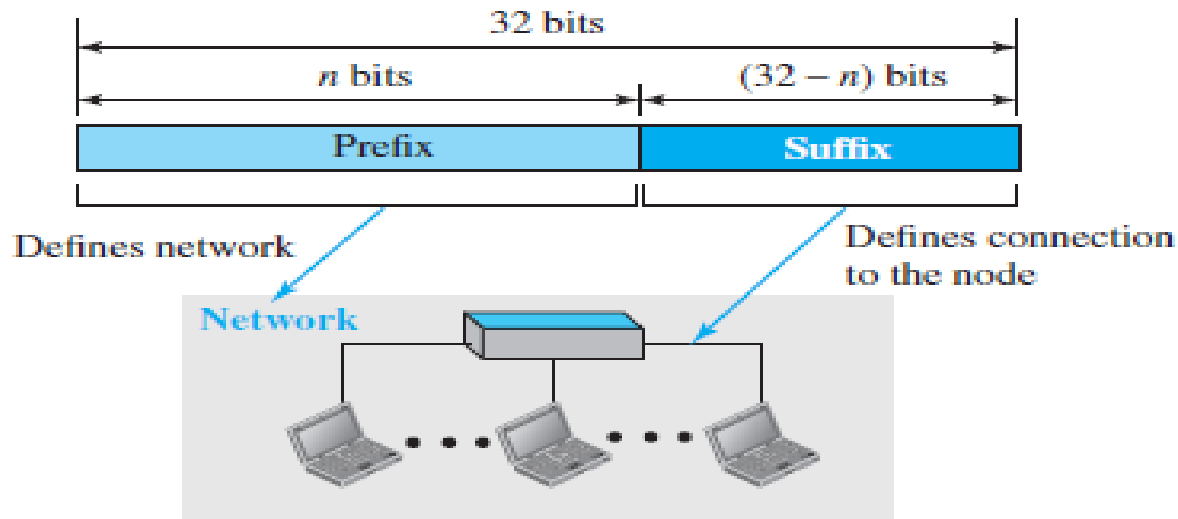
- a. *There must be no leading zero (045).*
- b. *There can be no more than four numbers.*
- c. *Each number needs to be less than or equal to 255.*
- d. *A mixture of binary notation and dotted-decimal notation is not allowed.*

Classful and Classless Addressing

Hierarchy in Addressing

- A 32-bit IPv4 address is also hierarchical, but divided only into two parts.
- The first part of the address, called the *prefix*, defines the network.
- The second part of the address, called the *suffix*, defines the node(host).

Hierarchy in addressing



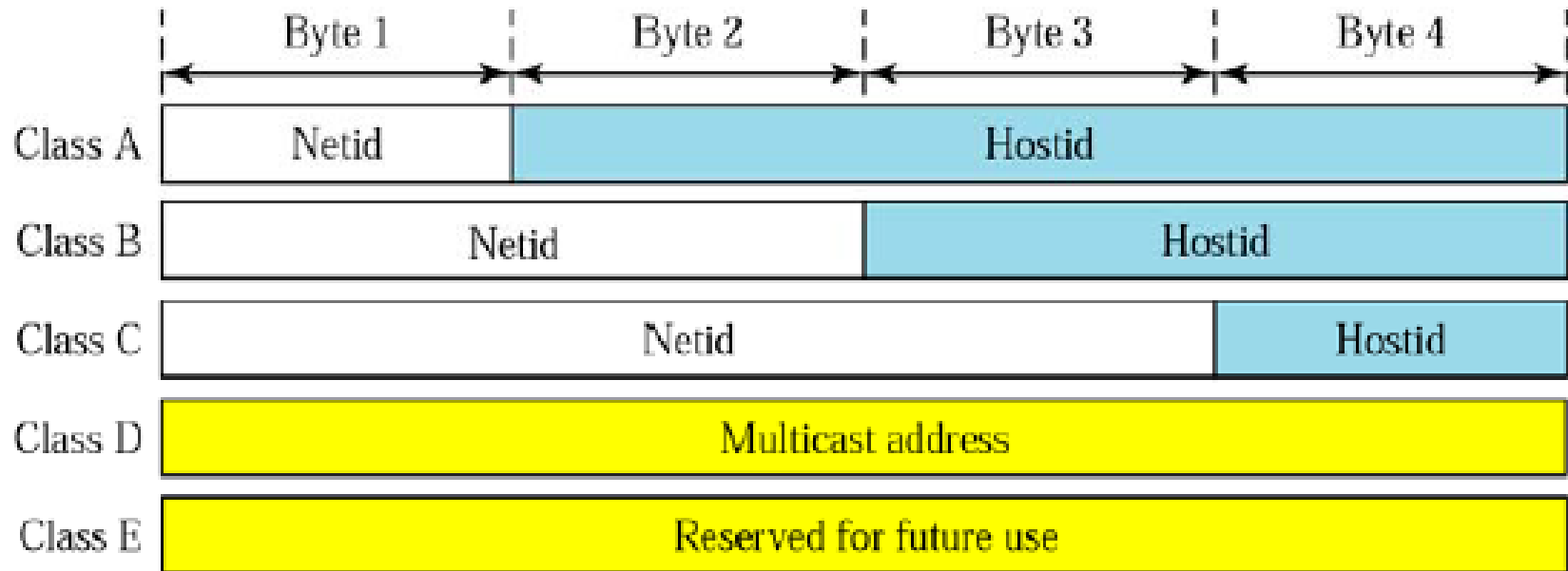


Classful Addressing

Note

**In classful addressing, the address space is divided into five classes:
A, B, C, D, and E.**

Classful Addressing



In classful addressing, the address space is divided into five classes: A, B, C, D, and E.

Finding the classes in binary and dotted-decimal notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation

00000000

01111111

10000000

10111111

11000000

11011111

11100000

11101111

11110000

11111111



Example 19.4

Find the class of each address.

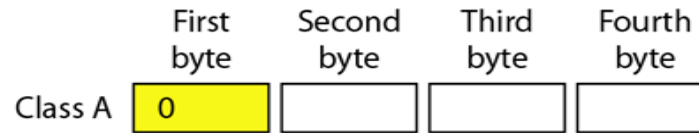
- a. 00000001 00001011 00001011 11101111
- b. 10100001 10000011 00011011 11111111
- c. 14.23.120.8
- d. 252.5.15.111

IPv4 Classes

Classes	Number of Networks <i>or</i> Number of blocks	Number of hosts per network <i>or</i> Block size	Range	Application
Class A	$2^7 = 128$ networks	$2^{24} = 16,777,216$ hosts	1.0.0.0 to 127.255.255.255	Unicast
Class B	$2^{14} = 16,384$ networks	$2^{16} = 65,536$ hosts	128.0.0.0 to 191.255.255.255	Unicast
Class C	$2^{21} = 2,097,152$ networks	$2^8 = 256$ hosts	192.0.0.0 to 223.255.255.255	Unicast
Class D	Multicast addresses 1	$2^{28} = 26,843,545$	224.0.0.0 to 239.255.255.255	Multicast
Class E	Future use 1	$2^{28} = 26,843,545$	240.0.0.0 to 255.255.255.255	Reserved

One problem is that each class is divided into fixed number of blocks with each block having a fixed size.

Example of Class A Network



Classes	Number of Networks	Number of hosts per network	Range	Application
Class A	$2^7 = 128$ networks	$2^{24} = 16777216$ hosts	1.0.0.0 to 127.255.255.255	Unicast

0.0.0.0
1.0.0.0
2.0.0.0
3.0.0.0

1.0.0.0 --- 1.0.0.1, 1.0.0.2, 1.0.0.3,... 1.0.1.0, 1.0.2.0...1.1.0.0,1.2.0.0... → 2^{24} addresses are possible

Network ID's (128 Networks)

102.0.0.0

- 102.168.212.226. Here, "102" helps you identify the network and 168.212.226 identify the host.

127.0.0.0

- 168.212.226.204, where *168 212* identifies the network and *226.204* helps you identify the Host network host.

Classes and Blocks

- **Class A address: designed for large organizations with a large** number of attached hosts or routers. (NASA , Pentagon etc.) (wasted and not used)
- **Class B address: designed for midsize organizations with ten** of thousands of attached hosts or routers(too large for many organizations)
- **Class C address: designed for small organizations with a** small number of attached hosts or routers.(too small for many organizations)
- **Class D address: designed for multicasting.** (waste of addresses)
- **Class E address: reserved for future use (waste of addresses)**

Classful Addressing: Classes and Blocks

Address Depletion : IPV4 Address space: 2^{32}

Increasing internet connected devices

- Near depletion of the available address because of the fast growth of the Internet.

Inefficient address use

- class A: This class can be assigned to only 128 organizations in the world, but each organization needs to have a single network with 16,777,216 nodes. Since there may be only a few organizations that are this large, most of the addresses in this class were wasted (unused).
- Class B addresses were designed for midsize organizations, but many of the addresses in this class also remained unused.
- Classes C block is too small for most mid size organizations.
- Class E addresses were almost never used, wasting the whole class.
 - **Solution:** To alleviate address depletion, two strategies were proposed and, to some extent, implemented: subnetting and supernetting
 - **Classless addressing**

Classful Addressing: Classes and Blocks

- **Subnet** :**Subnetting** is the practice of **dividing** a network into two or more smaller networks.
 - If an organization was granted a large block in classes A or B
 - It could divide the addresses into several contiguous groups and assign each group to smaller networks(**subnets**)
 - large organizations were not happy about dividing the block and giving some of the unused addresses to smaller organizations.

IP Address

Class	Starting Address	Ending Address	Subnet mask
A	0.0.0.0	127.255.255.255	255.0.0.0
B	128.0.0.0	191.255.255.255	255.255.0.0
C	192.0.0.0	223.255.255.255	255.255.255.0
D	224.0.0.0	239.255.255.255	255.255.255.255
E	240.0.0.0	255.255.255.255	255.255.255.255

A network mask helps you know which portion of the address identifies the network and which portion of the address identifies the node.

Classful Addressing: Classes and Blocks

- ***Supernetting :***

- Huge demand for midsize blocks.
- Although class A and B addresses are almost depleted, class C addresses are still available(size of block= 256 so address did not satisfy the needs).
- **In supernetting, an organization can combine** several class C blocks to create a larger range of Addresses
- Several networks are combined to create a supernetwork (**supernet**).



Advantages of Classful Addressing

Note

In classful addressing has many problem and became obsolete and but find class address easily

Drawbacks of Classfull addressing:

- The main flaw of classful addressing is **wastage of addresses** due to the **fixed size of blocks** having a **fixed number of addresses** in it.
- Lack of Internal Address Flexibility

Classless Addressing

- Classless Addressing is an improved IP Addressing system.
- It makes the allocation of IP Addresses more efficient.
- It replaces the older classful addressing system based on classes.
- It is also known as **Classless Inter Domain Routing (CIDR)**.

CIDR Block

- When a user asks for specific number of IP Addresses, CIDR dynamically assigns a block of IP Addresses based on certain rules.
- This block contains the required number of IP Addresses as demanded by the user.

Rules For Creating CIDR Block-

Rule-01: All the IP Addresses in the CIDR block must be contiguous.

Rule-02: The size of the block must be presentable as power of 2. Size of any CIDR block will always be in the form 2^1 , 2^2 , 2^3 , 2^4 , 2^5 and so on.

Rule-03: First IP Address of the block must be divisible by the size of the block.

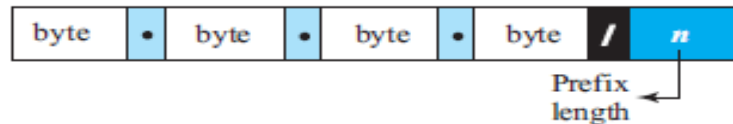
Example:

- 01100100.00000001.00000010.01000000 (100.1.2.64)
- It is divisible by 2^5 since its least significant 5 bits are zero.
- It is divisible by 2^6 since its least significant 6 bits are zero.
- It is not divisible by 2^7 since its least significant 7 bits are not zero.

CIDR Notation

18.20 *Slash notation (CIDR)*

a.b.c.d / n



Examples:

12.24.76.8/8

23.14.67.92/12

220.8.24.255/25

- They end with a slash followed by a number called as IP network prefix.
- /n indicates - number of bits used for the identification of network.
- Remaining bits are used for the identification of hosts in the network.

Example

182.0.1.2 / 28

- 28 bits are used for the identification of network.
- Remaining $32-28=4$ bits are used for the identification of host in the network

Given the CIDR representation 20.10.30.35 / 27. Find the range of IP Addresses in the CIDR block.

- 27 bits are used for the identification of network.
- Remaining 5 bits are used for the identification of hosts in the network.

20.10.30.35 / 27 (00010100.00001010.00011110.00100011 / 27)

- First IP Address = **00010100.00001010.00011110.001**00000 = 20.10.30.32
- Last IP Address = **00010100.00001010.00011110.001**11111 = 20.10.30.63
- Thus, Range of IP Addresses = [20.10.30.32 , 20.10.30.63] =
- $2^{32-27}=32$ IP addresses

Given the CIDR representation 100.1.2.35 / 20. Find the range of IP Addresses in the CIDR block.

- 20 bits are used for the identification of network.
- Remaining 12 bits are used for the identification of hosts in the network.

100.1.2.35 / 20 (01100100.00000001.00000010.00100011 / 20)

- First IP Address = **01100100.00000001.0000**0000.00000000 = 100.1.0.0
- Last IP Address = **01100100.00000001.0000**1111.11111111 = 100.1.15.255
- Thus, Range of IP Addresses = [100.1.0.0 , 100.1.15.255]
- $2^{32-20} = 4096$ IP addresses

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. Find the range of IP Addresses in the CIDR block.

11001101. 0001 0000. 00100101. 00100111 (205.16.37.39/28)

First address

11001101. 0001 0000. 00100101. 00100000 (205.16.37.32)

Last address

11001101. 0001 0000. 00100101. 00101111 (205.16.37.47)

Number of IP addresses: $2^{32-28} = 2^4 = 16$

Other Method

CIDR Block: 205.16.37.39 /28

Address:	11001101	00010000	00100101	00100111
Mask:	11111111	11111111	11111111	11110000
First address:	11001101	00010000	00100101	00100000

AND

Address:	11001101	00010000	00100101	00100111
Mask complement:	00000000	00000000	00000000	00001111
Last address:	11001101	00010000	00100101	00101111

OR

Consider a block of IP Addresses ranging from 150.10.20.64 to 150.10.20.127.

Is it a CIDR block?

If yes, give the CIDR representation.

- According to Rule-01, all the IP Addresses must be contiguous. Rule-01 is satisfied.
- According to Rule-02, size of the block must be presentable as 2^n .
 - Number of IP Addresses in given block = $127 - 64 + 1 = 64$.
 - Size of the block = 64 which can be represented as 2^6 .
- According to Rule-03, first IP Address must be divisible by size of the block. 150.10.20.64 = 150.10.20.01**000000** is divisible by 2^6 since its 6 least significant bits are zero. Rule-03 is satisfied
- Since all the rules are satisfied, therefore given block is a CIDR block.
- To have 2^6 total number of IP Addresses, 6 bits are required in the Host ID part. So, Number of bits in the Network ID part = $32 - 6 = 26$.
- 150.10.20.64 / 26

Example 18.4

- An ISP has requested a block of 1000 addresses. Since 1000 is not a power of 2, 1024 addresses are granted. How to calculate the prefix length.
- $2^{10}=1024$
- The **prefix length** is **calculated** as $n = 32 - \log_2 1024 = 22$ so it is /22
- The granted block is 18.14.12.0/22. The first address is 18.14.12.0/22 and the last address is 18.14.15.255/22.

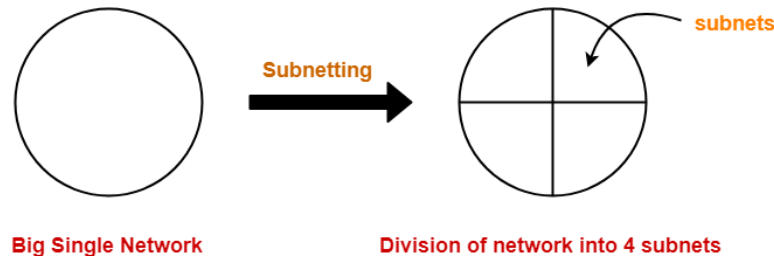
Example to demonstrate the address belongs to many blocks

- In classless addressing, an address cannot per se define the block the address belongs to. For example, the address **230.8.24.56** can belong to many blocks. Some of them are shown below with the value of the prefix associated with that block.

Prefix length:16	→	Block:	230.8.0.0	to	230.8.255.255
Prefix length:20	→	Block:	230.8.16.0	to	230.8.31.255
Prefix length:26	→	Block:	230.8.24.0	to	230.8.24.63
Prefix length:27	→	Block:	230.8.24.32	to	230.8.24.63
Prefix length:29	→	Block:	230.8.24.56	to	230.8.24.63
Prefix length:31	→	Block:	230.8.24.56	to	230.8.24.57

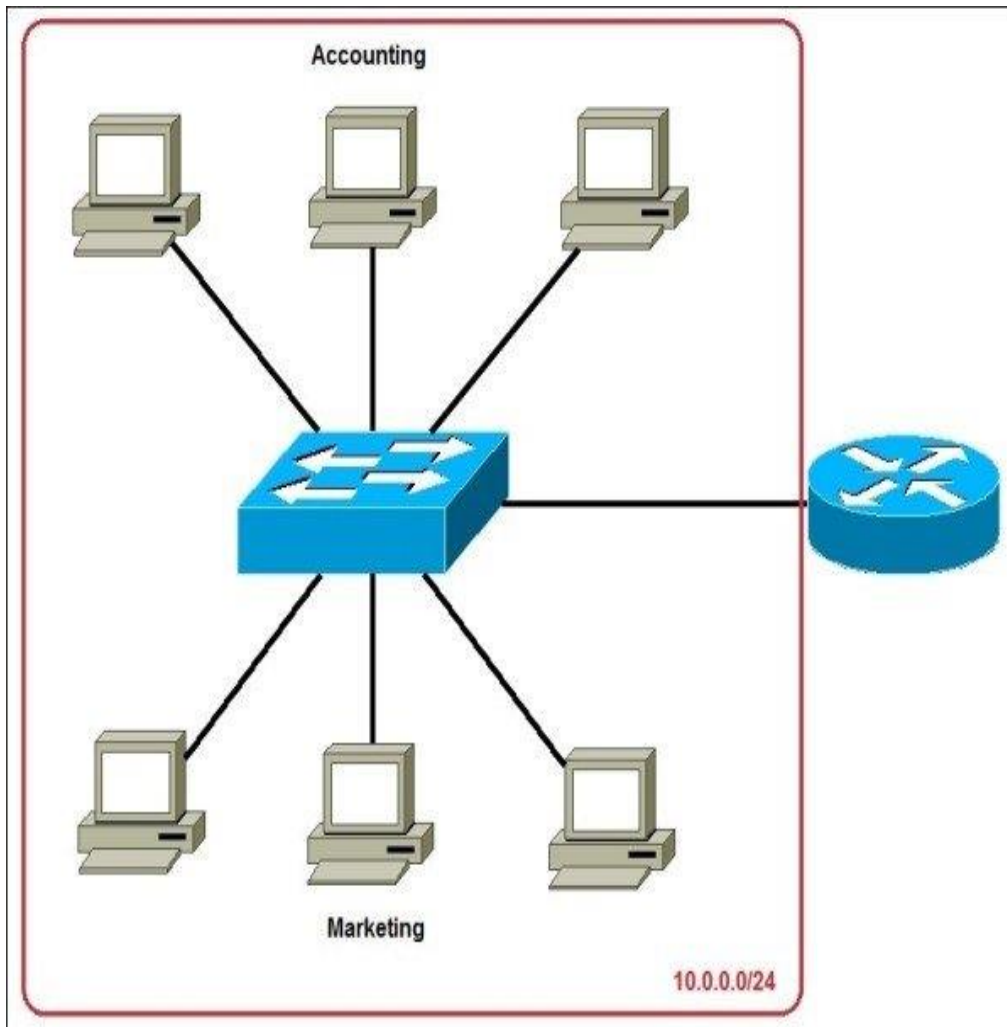
Subnetting

- **Subnetting** is the process of dividing a single network into two or more smaller(sub) networks.



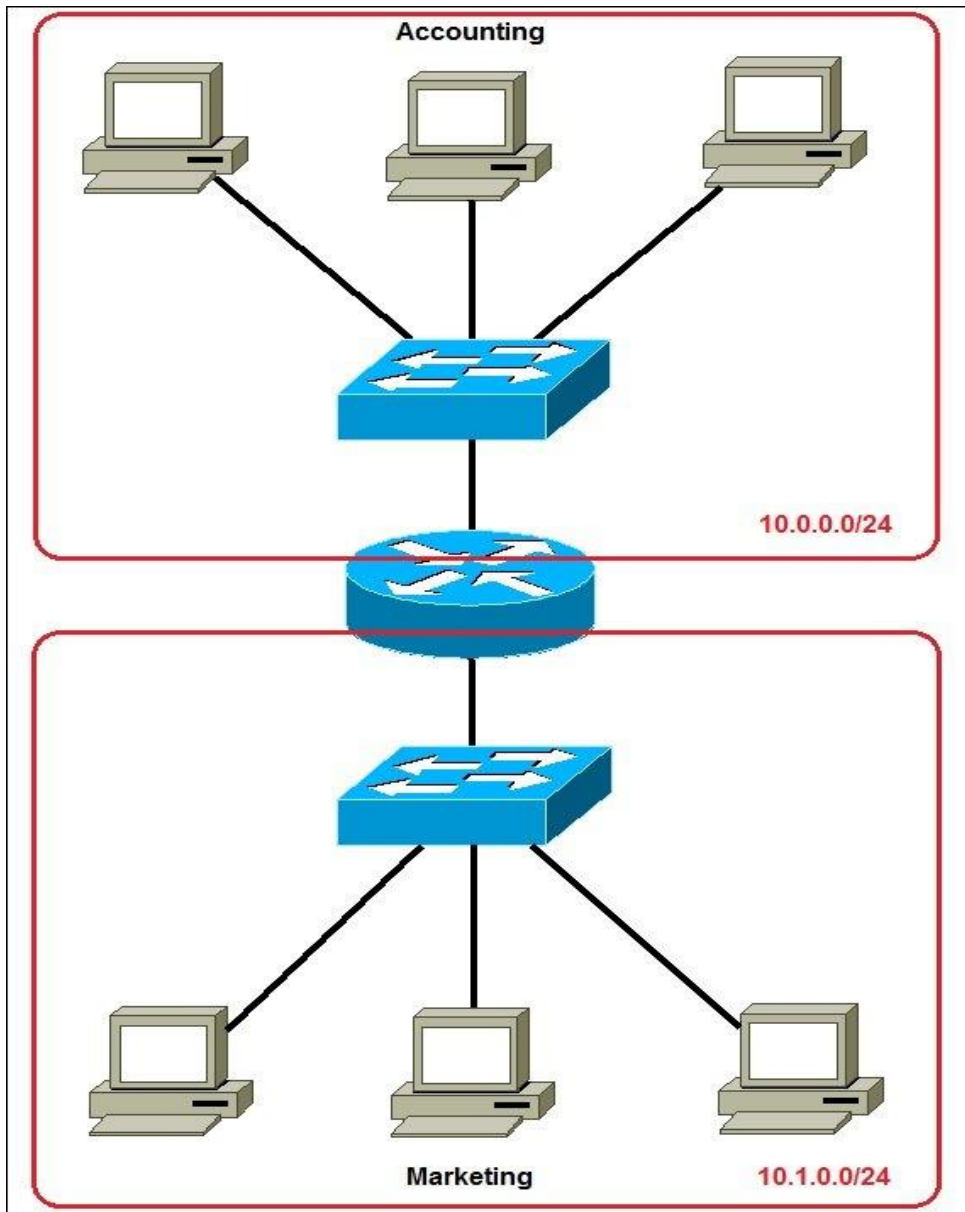
- Subnetting is the process of **stealing/borrow bits from the HOST part of an IP address** in order to divide the larger network into smaller sub-networks called subnets. The number of bits borrowed depends on the number of subnets created
- Each subnet has its unique network address known as its subnet ID
- **Objectives of Subnetting**
 - To avoid the wastage of IP address
 - It increases routing efficiency
 - Enhances the security of the network
 - Reduces the size of the broadcast domain.
 - To relieve network congestion

Why Subnetting ?



- In the picture above we have one huge network: **10.0.0.0/24**. All hosts on the network are in the same subnet, which has the following disadvantages:
- **a single broadcast domain** – all hosts are in the same broadcast domain. A broadcast sent by any device on the network will be processed by all hosts, creating lots of unnecessary traffic.
- **network security** – each device can reach any other device on the network, which can present security problems. For example, a server containing sensitive information shouldn't be in the same network as user's workstations.
- **organizational problems** – in a large networks, different departments are usually grouped into different subnets. For example, you can group all devices from the **Accounting** department in the same subnet and then give access to sensitive financial data only to hosts from that subnet.

Why Subnetting ?



- Now, two subnets were created for different departments:
- **10.0.0.0/24** for Accounting and **10.1.0.0/24** for Marketing. Devices in each subnet are now in a different broadcast domain. This will reduce the amount of traffic flowing on the network and allow us to implement packet filtering on the router.

Subnets and Subnet Masks

A block of addresses to be split into several parts for internal use as multiple networks, while still acting like a single network to the outside world. This is called subnetting and the networks (such as Ethernet LANs) that result from dividing up a larger network are called **subnets**.

Table 19.2 *Default masks for classful addressing*

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

Designing Subnets

Assumption

- Total number of addresses granted to the organization is N .
- Prefix length is n .
- Assigned number of addresses to each sub network is N_{sub} .
- Prefix length for each sub network is n_{sub} .

Rules

- Number of addresses in each subnetwork should be a power of 2.
- prefix length for each subnetwork should be found using the following formula:
 - $n_{sub} = 32 - \log_2 N_{sub}$
- The starting address in each subnetwork should be divisible by the number of addresses in that subnetwork. This can be achieved if we first assign addresses to larger subnetworks.

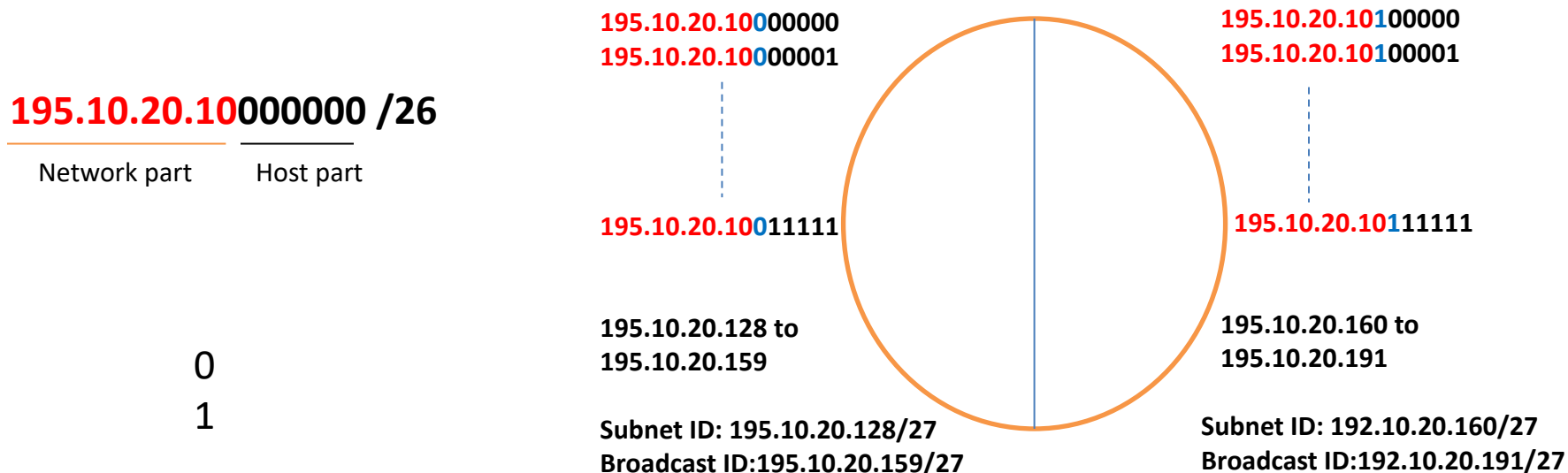
1)An organization is granted the block 195.10.20.128/26. The admin wants to create 2 subnets.

Step 1:Bits to be barrowed to cerate 2 subnets $2^1 = 2$ subnets

Step 2:Number of IP addresses possible $2^{32-26} = 2^6 = 64$ IP addresses

Step 3:Number of IP addresses in each subnet $64/2 = 32$ IP addresses

Step 4:Subnet prefix for each subnet $n1=32 - \log_2 32 = 32-5=27$ i.e. /27



2)An organization is granted the block 195.10.20.128/26. The admin wants to create 4 subnets.

Step 1:Bits to be barrowed to cerate 4 subnets $2^2 = 4$ subnets

Step 2:Number of IP addresses possible $2^{32-26} = 2^6 = 64$ IP addresses

Step 3:Number of IP addresses in each subnet $64/4 = 16$ IP addresses

Step 4:Subnet prefix for each subnet $n1 = 32 - \log_2 16 = 32 - 4 = 28$ i.e. /28

Subnet1: 195.10.20.10000000

Subnet 2: 195.10.20.10010000

Subnet 3: 195.10.20.10100000

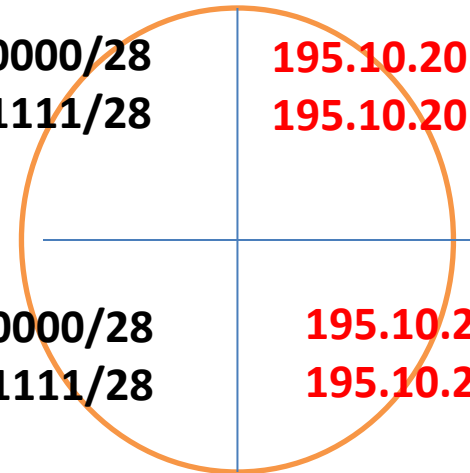
Subnet 4: 195.10.20.10110000

195.10.20.10000000/28
195.10.20.10001111/28

195.10.20.10010000
195.10.20.10011111

195.10.20.10100000/28
195.10.20.10101111/28

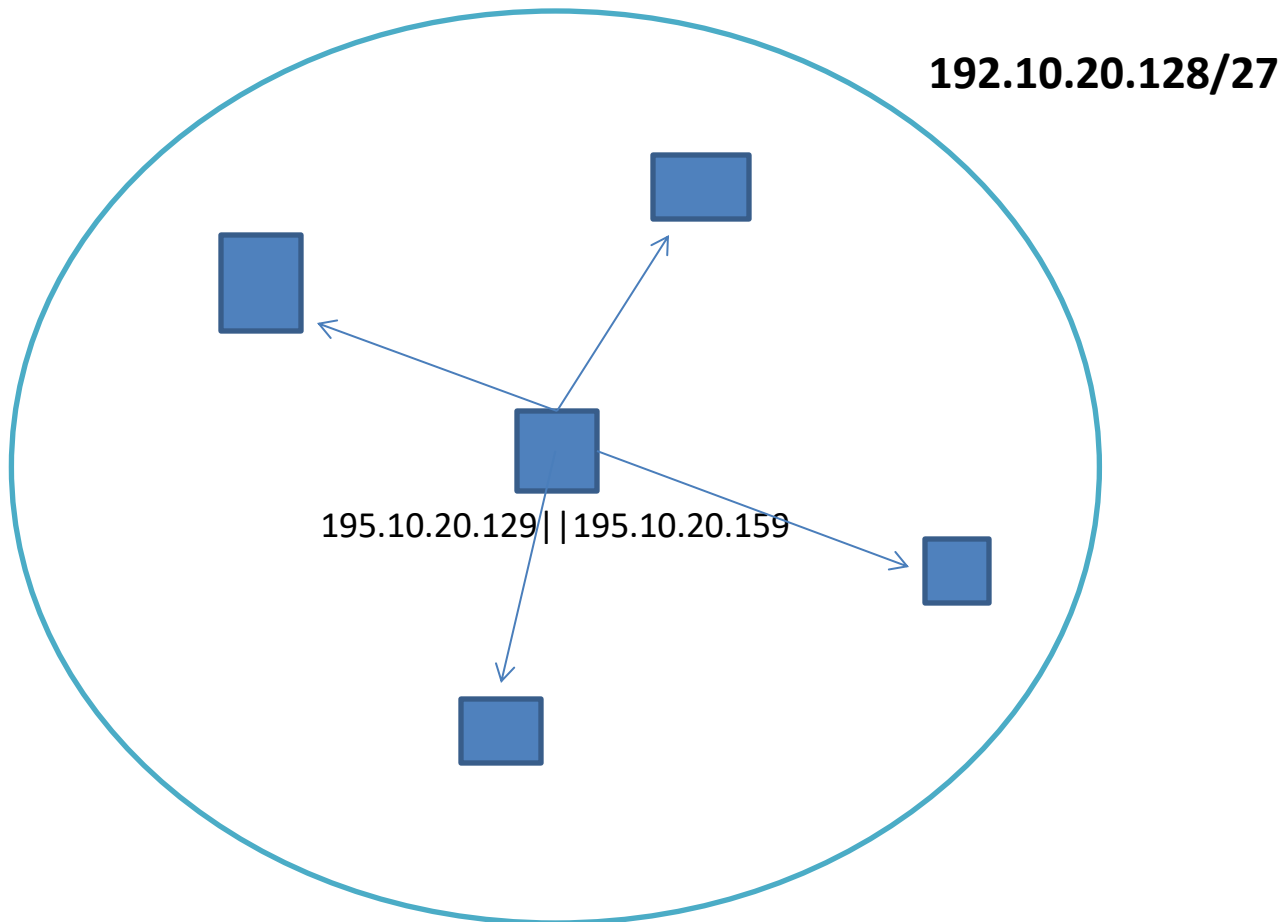
195.10.20.10110000
195.10.20.10111111



Broadcast ID: An **address** that enables transmission to every node in a local **network**.

Subnet ID: 195.10.20.128/27

Broadcast ID: 195.10.20.159/27



3)An organization is granted the block 195.10.20.128/26. The admin wants to create 8 subnets.

4)An organization is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets.

- 1.Find the subnet prefix
- 2.Find the number of addresses in each subnet
- 3.Find the first and the last address in the first subnet (1)
- 4.Find the first and the last address in the last subnet (32)

Step 1:Bits to be borrowed to create 32 subnets $2^{\boxed{5}} = 32$ subnets

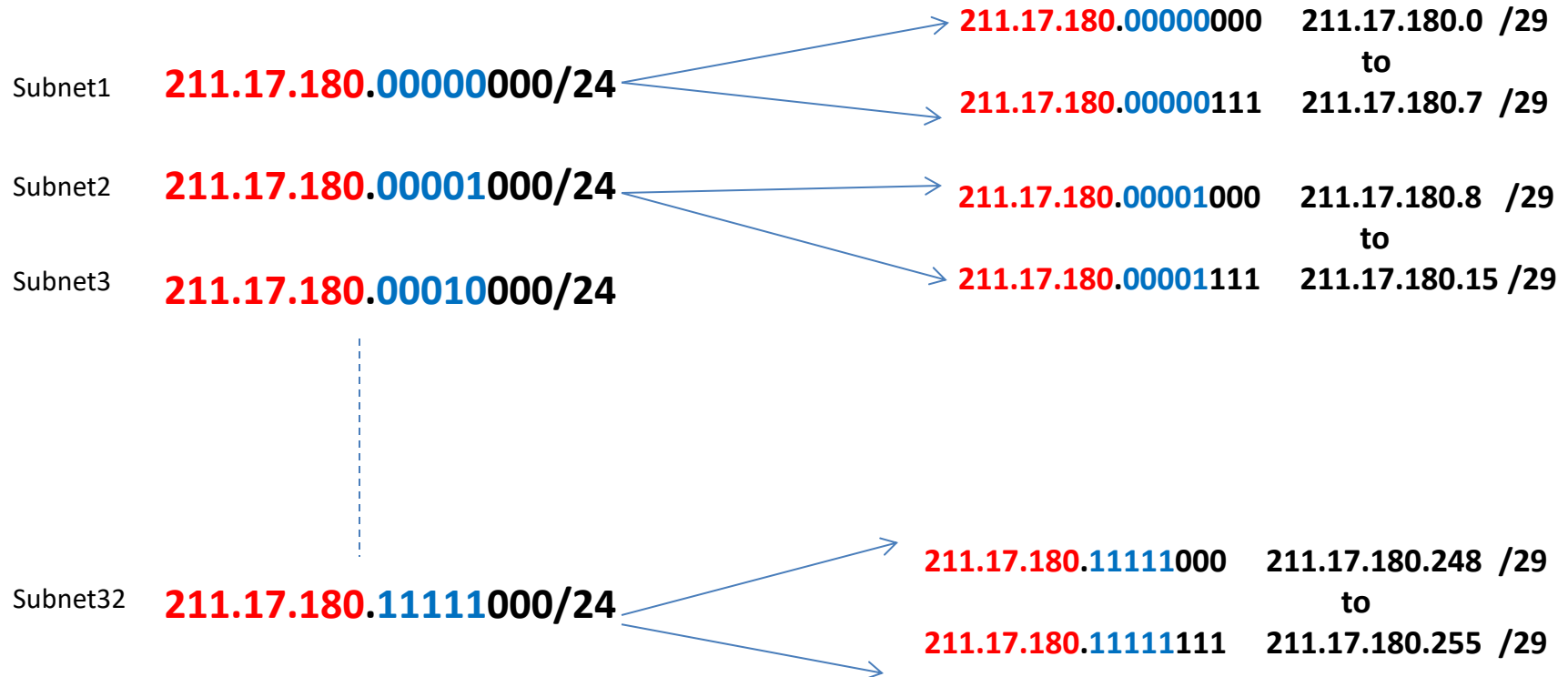
Step 2:Number of IP addresses possible $2^{\boxed{32-24}} = 2^{\boxed{8}} = 256$ IP addresses

Step 3:Number of IP addresses in each subnet $256/32 = 8$ IP addresses

Step 4:Subnet prefix for each subnet $n1=32 - \log_2 8 = 32-3=29$ i.e. /29

211.17.180.00000000/24

Network part Host part



5)An organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets.

Find the subnet prefix.

Find the number of addresses in each subnet.

Find the first and the last address in the first subnet.

5)An organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets.

Find the subnet prefix.

Find the number of addresses in each subnet.

Find the first and the last address in the first subnet.

Step 1:Bits to be barrowed to cerate 1024 subnets $2^{\boxed{10}} = 1024$ subnets

Step 2:Number of IP addresses possible $2^{\boxed{32-16}} = 2^{\boxed{16}} = 65536$ IP addresses

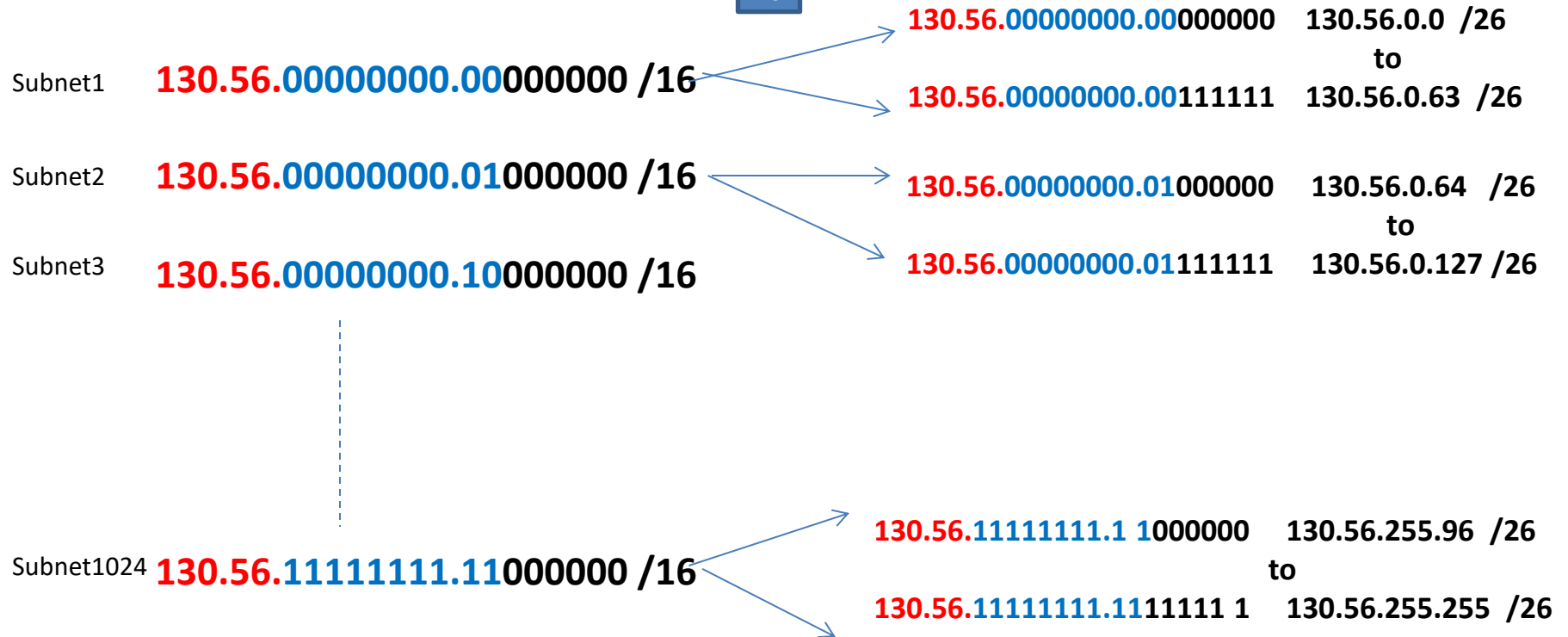
Step 3:Number of IP addresses in each subnet $65536/1024 = 64$ IP addresses

Step 4:Subnet prefix for each subnet $n1=32 - \log_2 64 = 32-6=26$ i.e. /26

130.56.0.0 /16

Network part Host part

10



4)An organization is granted a block of addresses with the beginning address 14.24.74.0/24. The organization needs to have 3 subblocks of addresses to use in its three subnets: one subblock of 10 addresses, one subblock of 60 addresses, and one subblock of 120 addresses. Design the subblocks.

- Number of IP addresses possible $2^{32-24} = 2^8 = 256$ IP addresses

14.24.74.0 /24
Network part Host part

- The first address is 14.24.74.0/24 and the last address is 14.24.74.255/24

4)An organization is granted a block of addresses with the beginning address 14.24.74.0/24. The organization needs to have 3 subblocks of addresses to use in its three subnets: one subblock of 10 addresses, one subblock of 60 addresses, and one subblock of 120 addresses. Design the subblocks.

1)120 addresses.

It is not power of 2. We can allocate 128 addresses.

Subnet mask: $n1=32 - \log_2 128 = 32-7=25$ i.e. /25

14.24.74.00000000 14.24.74.0 /25

14.24.74.01111111 14.24.74.127 /25

2)60 addresses.

It is not power of 2. We can allocate 64 addresses.

Subnet mask: $n1=32 - \log_2 64 = 32-6=26$ i.e. /26

14.24.74.10000000 14.24.74.128 /26

14.24.74.10111111 14.24.74.191 /26

3)10 addresses.

It is not power of 2. We can allocate 16 addresses.

Subnet mask: $n1=32 - \log_2 16 = 32-4=28$ i.e. /28

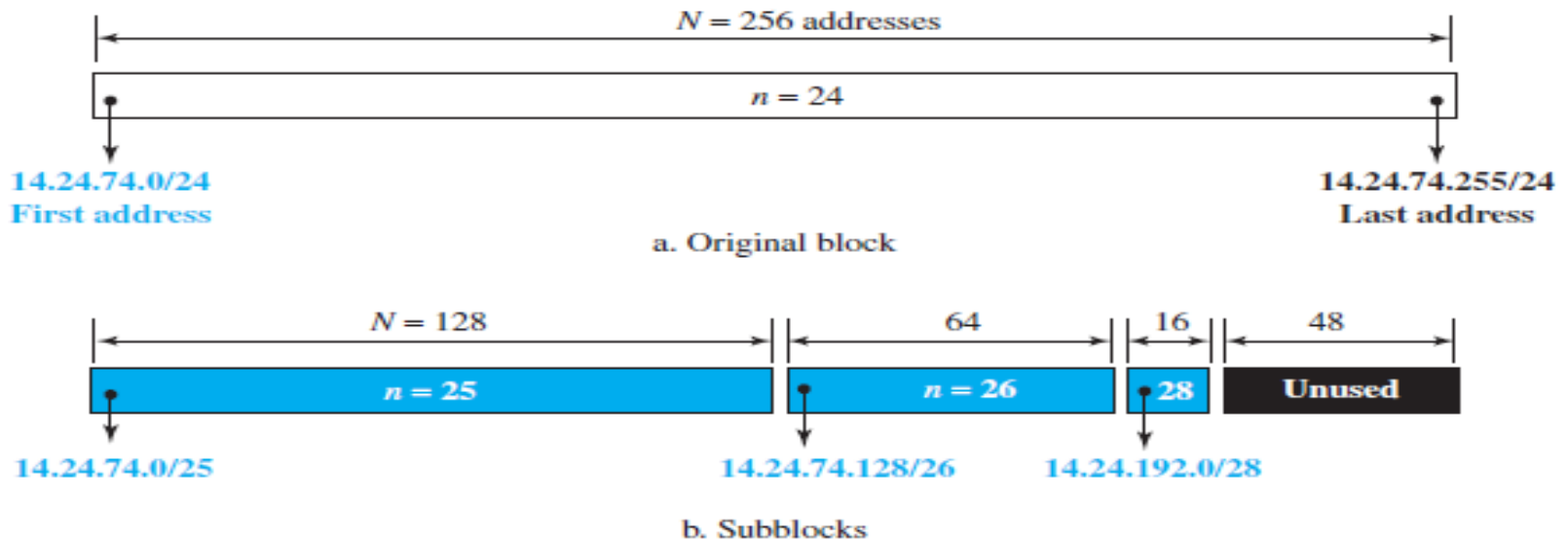
14.24.74.11000000 14.24.74.192 /28

14.24.74.11001111 14.24.74.207 /28

Analysis

- The result is $128+64+16=208$ addresses, which means $256-208=48$ addresses are left in reserve. The first address in this range is 14.24.74.208. The last address is 14.24.74.255.

Figure 18.23 *Solution to Example 18.5*



An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- i. The first group has 64 customers each needs 256 addresses.**
- ii. The second group has 128 customers, each needs 128 addresses.**
- iii. The third group has 128 customers, each needs 64 addresses.**

Design the sub-blocks and find out how many addresses are still available after these allocations.

Assignment

An isp is granted the block 80.70.56.0/21. The ISP needs to allocate addresses for two organizations each with 500 addresses, two organizations each with 250 addresses, and three organizations each with 50 addresses.

- A. Find the number and range of addresses in the isp block. $2^{32-21}=2048$ Ip address
- B. Find the range of addresses for each organization and the range of unallocated addresses.

500 IP address – 2 organizations
250 IP address – 2 organizations
50 Ip address – 3 organizations

Assignment

An isp is granted the block 80.70.56.0/21. The isp needs to allocate addresses for two organizations each with 500 addresses, two organizations each with 250 addresses, and three organizations each with 50 addresses.

- A. Find the number and range of addresses in the isp block.
- B. Find the range of addresses for each organization and the range of unallocated addresses.

A. No. of addresses= 2^{11}

Range of addresses= 80.70.56.0/21 -80.70.63.255/21

B. 80.70.56.0/23 - 80.70.57.255/23 (512 addresses)

80.70.58.0/23 - 80.70.59.255/23 (512 addresses)

80.70.60.0/24 - 80.70.60.255/24 (256 addresses)

80.70.61.0/24 - 80.70.61.255/24 (256 addresses)

80.70.62.0/26 -80.70.62.63/26 (64 addresses)

80.70.62.64/26- 80.70.62.127/26 (64 addresses)

80.70.62.128/26 - 80.70.62.191/26 (64 addresses)

addresses are always assigned in the power of 2.

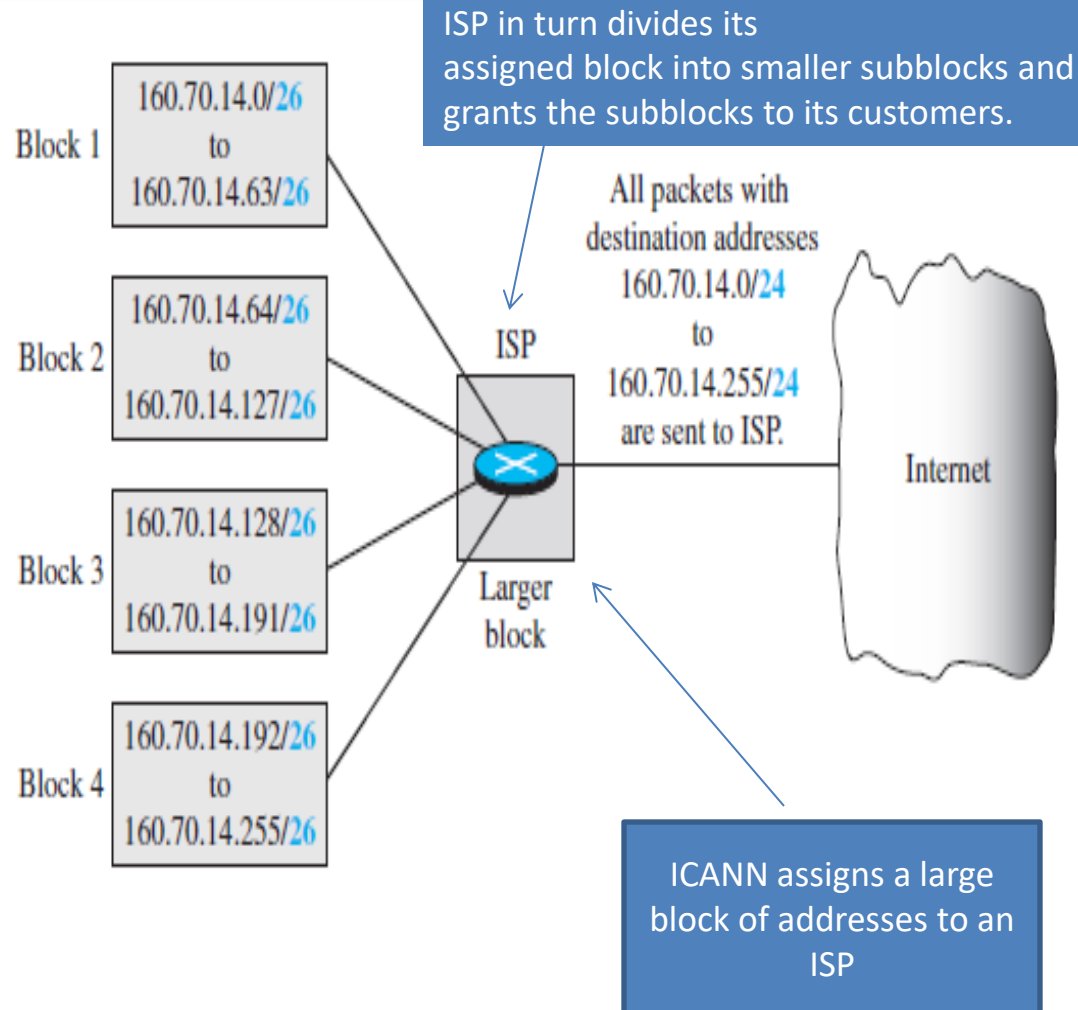
unallocated addresses :80.70.62.192/26 - 80.70.63.255/26

Address Aggregation (An advantage of CIDR strategy)

- **Address**

Aggregation: CIDR strategy is **address aggregation** (sometimes called *address summarization or route summarization*)

Figure 18.24 Example of address aggregation



Special Addresses

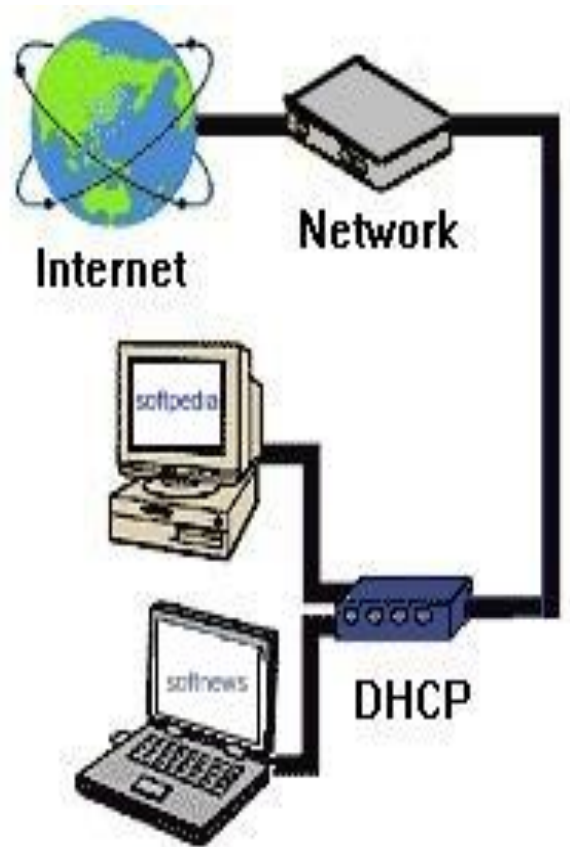
- Five special addresses that are used for special purposes.
 - *This-host address.*
 - *Limited-broadcast address.*
 - *Loopback address.*
 - *Private addresses.*
 - *Multicast addresses.*

- **this-host Address: 0.0.0.0/32**
 - It is used whenever a host needs to send an IP datagram but it does not know its own address to use as the source address (used in DHCP.)
- **Limited-broadcast Address: 255.255.255.255/32**
 - It is used whenever a router or a host needs to send a datagram to all devices in a network
- **Loopback Address: 127.0.0.0/8**
 - it will remain in the host
 - (Client/server program)
- **Private Addresses:**
 - Four blocks are assigned as private addresses:
 - **10.0.0.0/8 , 172.16.0.0/12, 192.168.0.0/16, and 169.254.0.0/16.** (Used in NAT)
- **Multicast Addresses:**
 - Block 224.0.0.0/4 is reserved for multicast addresses

HISTORY- DHCP (Dynamic Host Configuration Protocol)

- ❑ DHCP was created by the Dynamic Host Configuration Working Group of the Internet Engineering Task Force
- ❑ October 1993: RFC 1531 initially defined DHCP as a standard-track protocol succeeding the Bootstrap Protocol (BOOTP), which is a network protocol used by a network client **to obtain an IP address from a configuration server**
- ❑ October 1997: RFC 2131 released is the current DHCP definition for Internet Protocol version 4 (IPv4) networks
- ❑ The extensions of DHCP for IPv6 (DHCPv6) were published as RFC 3315

Dynamic Host Configuration Protocol (DHCP)



- Address are allocated automatically using the **Dynamic Host Configuration Protocol (DHCP)**.
- DHCP is often called a *plug and-play protocol*.
- DHCP to assign permanent IP addresses or also temporary, on demand, IP addresses to hosts or routers.

Motivation for DHCP

- Configuration parameters for network hosts
 - IP address
 - Router
 - Subnet Mask
 - Others..

Two types of IP Addresses

- DHCP is used to assign IP addresses to hosts or workstations on the network
- Two types of IP addresses:
 - ✓ Static
 - ❖ Is a number that is assigned to a computer by an Internet service provider (ISP) to be its permanent address on the Internet
 - ✓ Dynamic
 - ❖ The temporary IP address is called a dynamic IP address

Why is DHCP Important?

- Important when it comes to adding a machine to a network
- When computer requests an address, the administrator would have to manually configure the machine
 - ✓ Mistakes are easily made
 - ✓ Causes difficulty for both administrator as well as neighbors on the network
- DHCP solves all the hassle of manually adding a machine to a network

Advantages of DHCP

Application Layer Protocol – Automatically assign the IP address for the system

- DHCP minimizes the administrative burden
- By using DHCP there is no chance to conflict IP address
- By using DHCP relay agent you provide IP address to another network

Dynamic Host Configuration Protocol (DHCP)

Figure 18.25 *DHCP message format*

0	8	16	24	31
Opcode	Htype	HLen	HCount	
Transaction ID				
Time elapsed		Flags		
Client IP address				
Your IP address				
Server IP address				
Gateway IP address				
Client hardware address				
Server name				
Boot file name				
Options				

Fields:

Opcode: Operation code, request (1) or reply (2)

Htype: Hardware type (Ethernet, ...)

HLen: Length of hardware address

HCount: Maximum number of hops the packet can travel

Transaction ID: An integer set by the client and repeated by the server

Time elapsed: The number of seconds since the client started to boot

Flags: First bit defines unicast (0) or multicast (1); other 15 bits not used

Client IP address: Set to 0 if the client does not know it

Your IP address: The client IP address sent by the server

Server IP address: A broadcast IP address if client does not know it

Gateway IP address: The address of default router

Server name: A 64-byte domain name of the server

Boot file name: A 128-byte file name holding extra information

Options: A 64-byte field with dual purpose described in text

Message Fields

- **opcode**: Indicates a request or a reply
 - Request
 - Reply
- **HWtype**: The type of hardware, for example:
 - Ethernet
 - IEEE 802 networks
- **length**: Hardware address length in bytes. E.g., Ethernet and token-ring both use 6 bytes.
- **hops**: The client sets this to 0. It is incremented by a router that relays the request to another server and is used to identify loops. RFC 951 suggests that a value of 3 indicates a loop.

Contd.

- **Transaction ID:** A random number used to match this boot request with the response it generates.
- **Seconds:** Set by the client. It is the elapsed time in seconds since the client started its boot process.
- **Flags field:** First bit defines unicast (0) or multicast (1); other 15 bits not used

Contd.

- **Client IP address**: Set by the client. Either its known IP address, or 0.0.0.0.
- **Your IP address**: Set by the server if the client IP address field was 0.0.0.0.
- **Server IP address**: Set by the server.
- **Gateway IP address**: The address of default router.
- **Client hardware address**: Set by the client. DHCP defines a client identifier option that is used for client identification. If this option is not used the client is identified by its MAC address.

Contd.

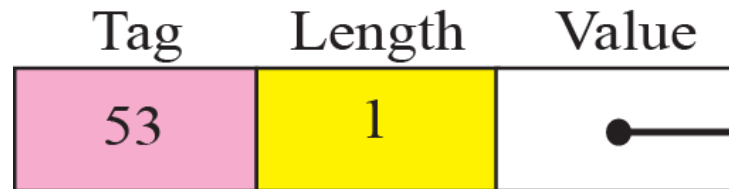
- **Server host name**: A 64-byte domain name of the server and Optional server host name terminated by X'00'.
- **Boot file name**: A 128-byte file name holding extra information . The value is terminated by X'00'.
- **Options**: A 64-byte field with dual purpose .

Subnet Mask, Name Server, Hostname, Domain Name, Forward On/Off, Default IP TTL, Broadcast Address, Static Route, Ethernet Encapsulation, X Window Manager, X Window Font, DHCP Msg Type, DHCP Renewal Time, DHCP Rebinding, Time SMTP-Server, SMTP-Server, Client FQDN, Printer Name, ...

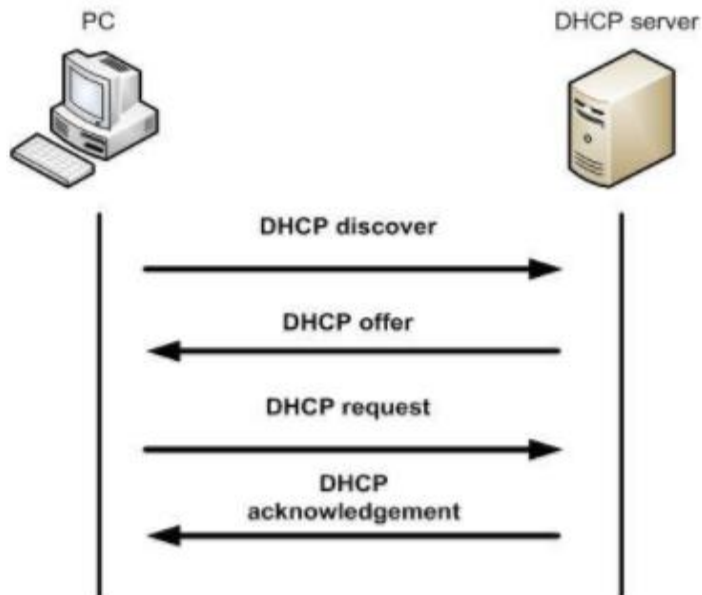
Options Field

- A 64-byte field with dual purpose.
 - Mainly to Carry additional information or specific vendor information.
 - **Magic cookie, in the format of an IP address with the value of 99.130.83.99.**
 - The next 60 bytes are options are :
 - 1-byte tag field
 - 1-byte length field,
 - Variable-length value field
- If the tag field is 53

DHCP Message Type



- 1 **DHCP**DISCOVER
- 2 **DHCP**OFFER
- 3 **DHCP**REQUEST
- 4 **DHCP**DECLINE
- 5 **DHCP**ACK
- 6 **DHCP**NACK
- 7 **DHCP**RELEASE
- 8 **DHCP**INFORM



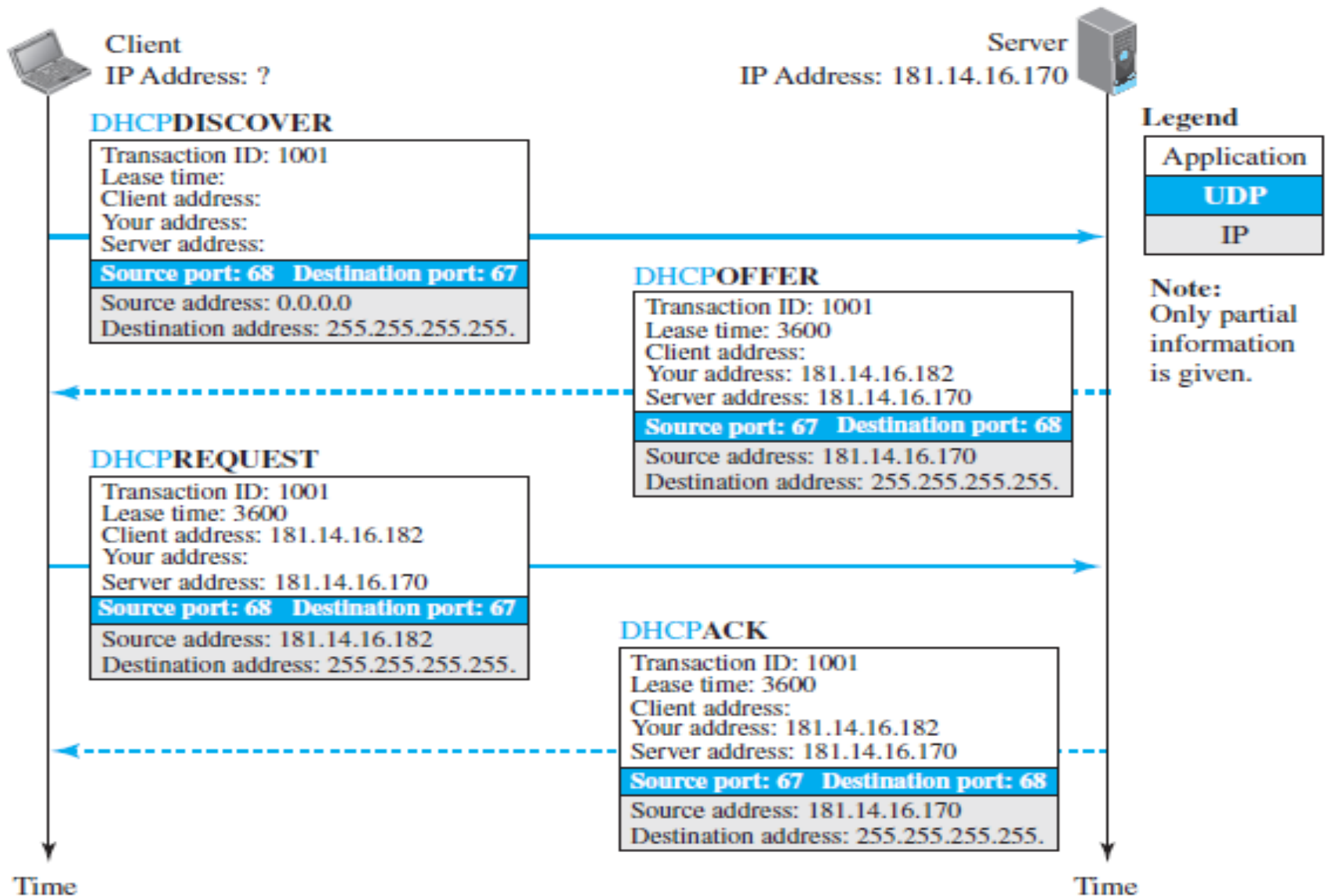
Message Types

- **DHCPDISCOVER**: Broadcast by a client to find available DHCP servers.
- **DHCPOFFER**: Response from a server to a DHCPDISCOVER and offering IP address and other parameters.
- **DHCPREQUEST**: Message from a client to servers that does one of the following:
 - Requests the parameters offered by one of the servers and declines all other offers.
 - Verifies a previously allocated address after a system or network change (a reboot for example).
 - Requests the extension of a lease on a particular address.

Contd.

- **DHCPACK**: Acknowledgement from server to client with parameters, including IP address.
- **DHCPNACK**: Negative acknowledgement from server to client, indicating that the client's lease has expired or that a requested IP address is incorrect.
- **DHCPDECLINE**: Message from client to server indicating that the offered address is already in use.
- **DHCPRELEASE**: Message from client to server canceling remainder of a lease and relinquishing network address.
- **DHCPINFORM**: Message from a client that already has an IP address (manually configured for example), requesting further configuration parameters from the DHCP server.

DHCP Operations



Client and server on the same network

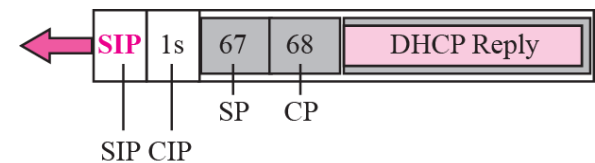
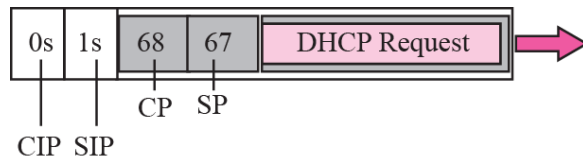
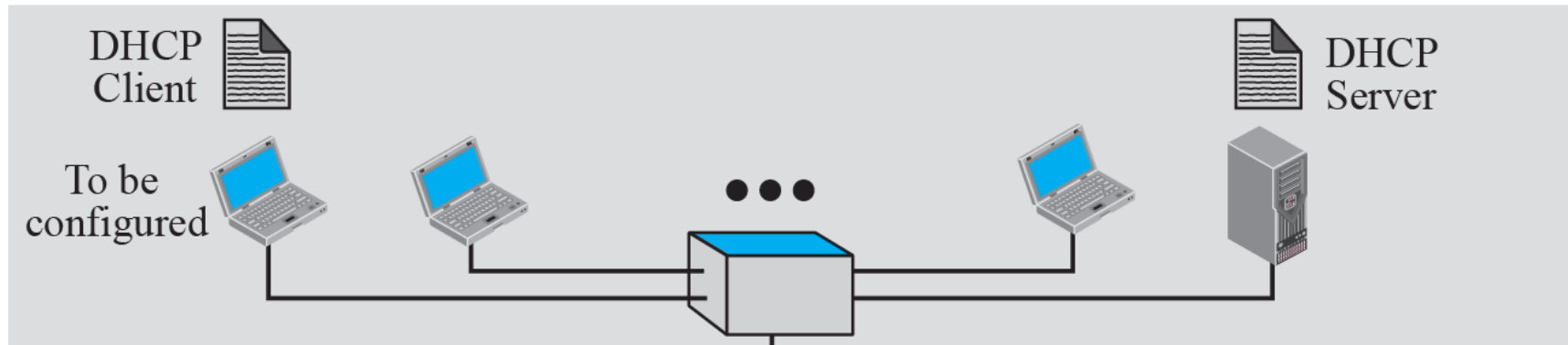
Legend

CP: Client Port Number

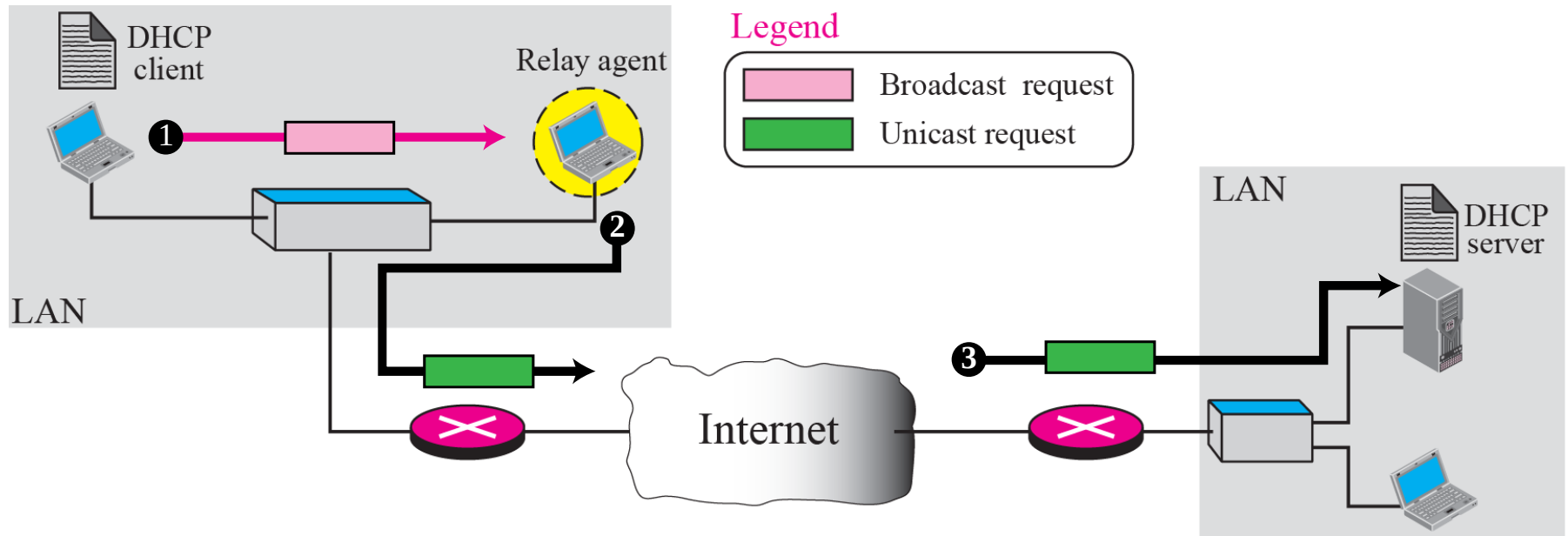
CIP: Client IP Address

SP: Server Port Number

SIP: Server IP Address



Client and server on two different networks



Note

Two Well-Known Ports

- DHCP uses two well-known ports (68 and 67) instead of one well-known and one ephemeral.
- Source Client is 68
- Destination port is 67

Error Control

- DHCP uses the service of UDP, which is not reliable.
- DHCP uses two strategies:
 - DHCP requires that UDP use the checksum.
 - DHCP client uses timers and a retransmission policy.
- DHCP forces the client to use a random number to set its timers.

Note

Port Number Range	Part Group
0 to 1023	Well Known (Contact) Ports
1024 to 49151	Registered Ports
49152 to 65535	Private and/or Dynamic Ports

Http

Client :3000

Server:80

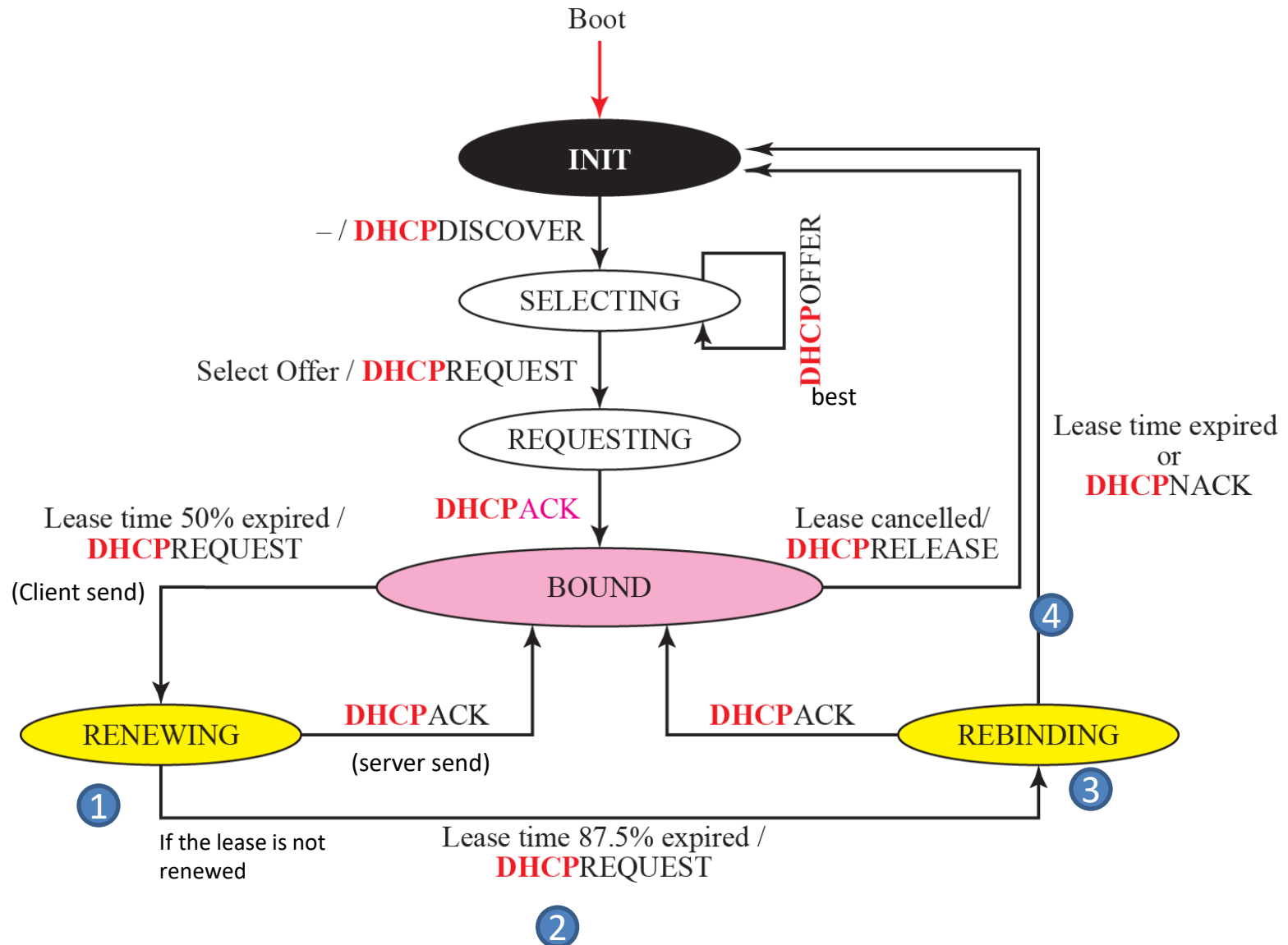
Two Well-Known Ports

- DHCP uses two well-known ports (68 and 67) instead of one well-known and one ephemeral (An **ephemeral port** is a short-lived **port** number used by an Internet Protocol (IP) transport protocol).
- Source Client is 68
- Destination port is 67
- Reason: response from the server to the client is broadcast (every host on the network)

Ex:

- DHCP client and a DAYTIME client - both waiting to receive a response from their corresponding server.
- Accidentally both are using the same temporary port number (56017, for example).
- The DHCP client processes the message; the DAYTIME client is totally confused with a strange message received. Using a well-known port number prevents this problem from happening.
- The response message from the DHCP server is not delivered to the DAYTIME client, which is running on the port number 56017, not 68.
- if two DHCP clients are running at the same time. The messages can be distinguished by the value of the transaction ID, which separates each response from the other.

Transition States



Timer at Client

- The client uses three timers:
 - *renewal timer (set to 50 percent of the lease time),*
 - *Rebinding timer (set to 75 percent of the lease time),*
 - *expiration timer (set to the lease time).*

Network Address Resolution (NAT)

- **Network Address Translation** (NAT) is designed for IP address conservation. NAT converts that local (private) IP address to a global (public) IP address.
- NAT operates on a **router**, usually connecting two networks together, and translates the private (not globally unique) addresses in the internal network into legal addresses, before packets are forwarded to another network.

Why NAT is required?

NAT provides the following benefits:

- Reuse of private IP addresses
- Enhancing security for private networks by keeping internal addressing private from the external network
- IP address spaces can be saved because many hosts connect to the global Internet by using a single dynamic external IP address.

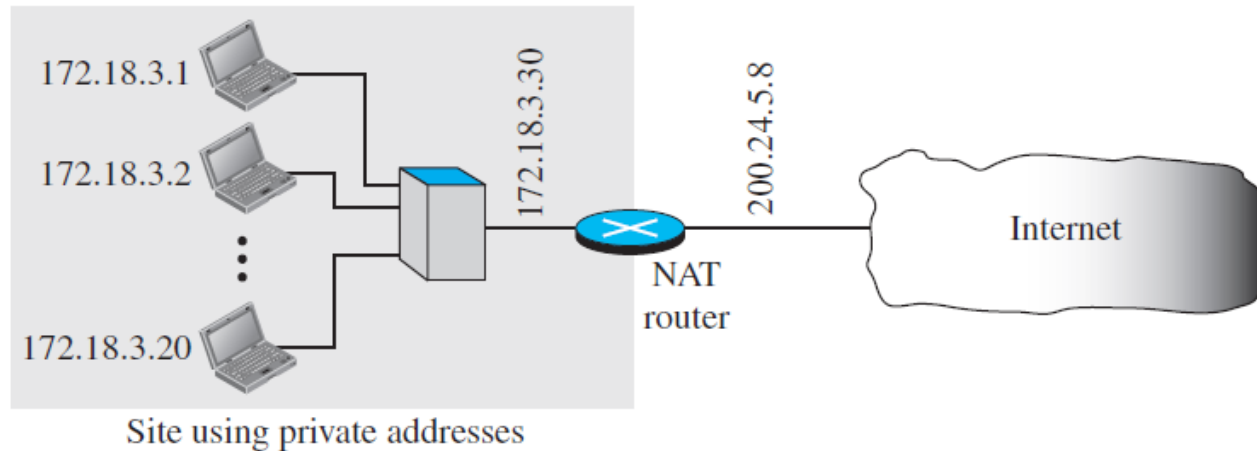
<https://www.youtube.com/watch?v=FTUV0t6JaDA>

<https://www.youtube.com/watch?v=QBqPzHEDzvo>

Network Address Translation

- A short term solution to the problem of the depletion of IP addresses.
 - Long term solution is IPv6
 - CIDR (Classless Interdomain Routing) has been a short term solution.
 - NAT is another short term solution
- NAT is a way to conserve IP addresses
 - **Hide a number of hosts behind a single IP address**
 - Use:
 - 10.0.0.0-10.255.255.255,
 - 172.16.0.0-172.32.255.255 or
 - 192.168.0.0-192.168.255.255 for local networks

Figure 18.29 NAT



The router that connects the network to the global address uses one private address and one global address. The private network is invisible to the rest of the Internet; the rest of the Internet sees only the NAT router with the address 200.24.5.8.

IANA has reserved:

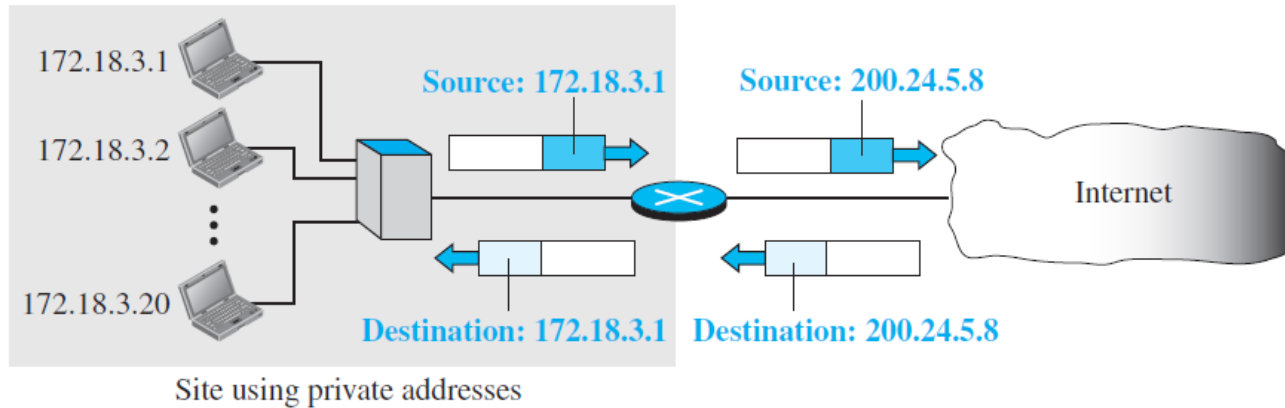
10.0.0.0 - 10.255.255.255 (10.0.0.0/8 prefix)

172.16.0.0 - 172.31.255.255 (172.16.0.0/12 prefix)

192.168.0.0 - 192.168.255.255 (192.168.0.0/16 prefix)

Address Translation

Figure 18.30 *Address translation*

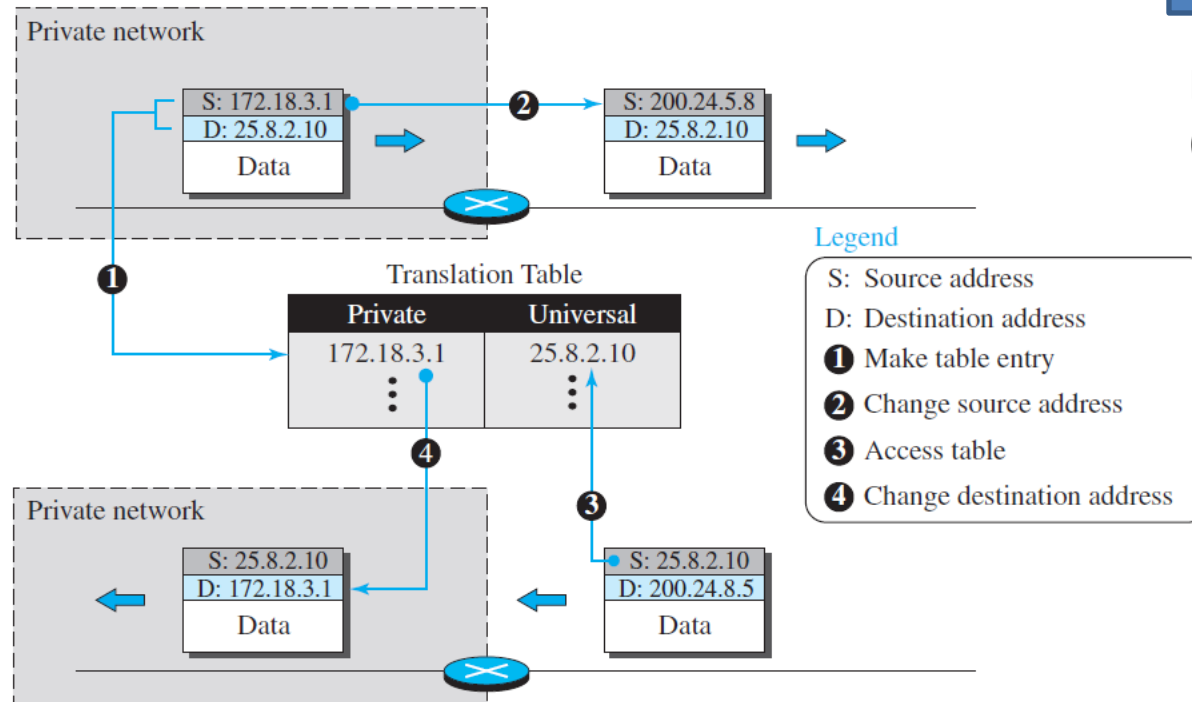


1. Outgoing packets go through the NAT router
2. All incoming packets also pass through the NAT router

Translation Table

Using One IP Address

Figure 18.31 Translation



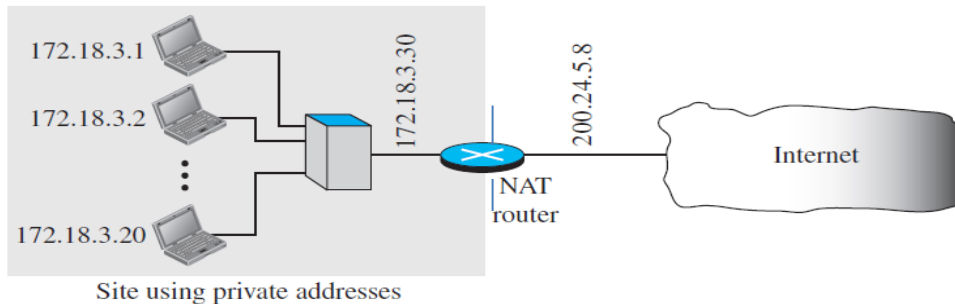
Facebook server
(25.8.2.10)

- A translation table has only two columns: the private address and the external address
- The router translates the source address of the outgoing packet, it also makes note of the destination address— where the packet is going
- When the response comes back from the destination, the router uses the source address of the packet to find the private address of the packet

Translation Table

Using a Pool of IP Addresses

Figure 18.29 NAT



25.8.2.10

Destination Server

Translation Table	
Private	Universal
172.18.3.1	25.8.2.10
⋮	⋮

- The use of only one global address by the NAT router allows only one private-network host to access a given external host.
- To remove this restriction, the NAT router can use a pool of global addresses. For example, instead of using only one global address (200.24.5.8), the NAT router can use four addresses (200.24.5.8, 200.24.5.9, 200.24.5.10, and 200.24.5.11).
- In this case, four private-network hosts can communicate with the same external host at the same time because each pair of addresses defines a separate connection.

Drawbacks

- No more than four connections can be made to the same destination.

Translation Table

Using Both IP Addresses and Port Addresses

Table 18.1 *Five-column translation table*

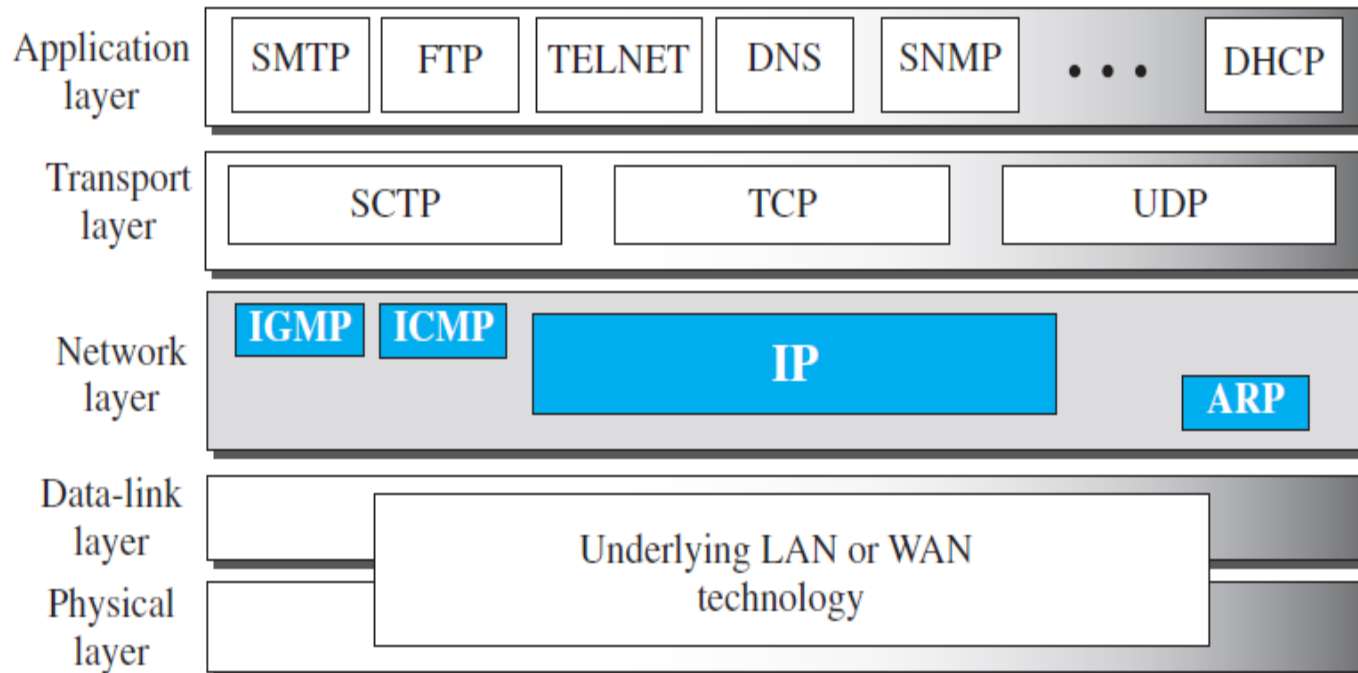
<i>Private address</i>	<i>Private port</i>	<i>External address</i>	<i>External port</i>	<i>Transport protocol</i>
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
172.18.3.1	3000	70.1.2.15	21	TCP

- To allow a many-to-many relationship between private-network hosts and external server programs, **we need more information in the translation table.**
- Suppose two hosts inside a private network with addresses 172.18.3.1 and 172.18.3.2 need to access the HTTP server on external host 25.8.3.2. If the translation table has five columns, instead of two, that include the source and destination port addresses and the transport-layer protocol, the ambiguity is eliminated.

INTERNET PROTOCOL (IP)

INTERNET PROTOCOL (IP)

Figure 19.1 *Position of IP and other network-layer protocols in TCP/IP protocol suite*



INTERNET PROTOCOL (IP)

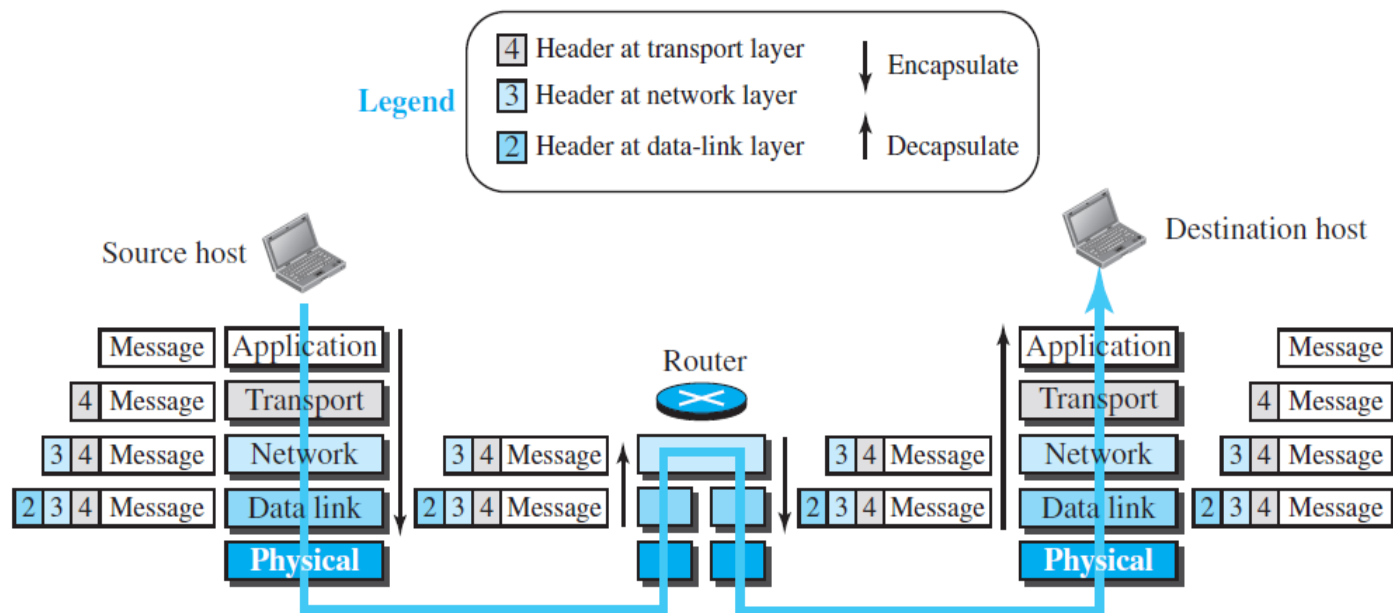
Protocols at Network Layer

- Internet Protocol version 4 (IPv4), is responsible for packetizing, forwarding, and delivery of a packet at the network layer.
- The Internet Control Message Protocol version 4 (ICMPv4) helps IPv4 to handle some errors that may occur in the network-layer delivery.
- The Internet Group Management Protocol (IGMP) is used to help IPv4 in multicasting
- The Address Resolution Protocol (ARP) is used to glue the network and data-link layers in mapping network-layer addresses to link-layer addresses.

IP

- IP is a **connection-less, unreliable protocol** (it does not guarantee the delivery of a datagram to its destination).
 - connection-less**: each datagram is handled independently, and each datagram can follow a different route to the destination. This implies that datagrams sent by the same source to the same destination could arrive out of order
- **IP provides *best effort services*** (does not provide full reliability) in the sense
 - There is no guarantee of delivery of error-free packets
 - There is no guarantee of ordered delivery of packets
 - *IPv4 packets can be corrupted, be lost, or be delayed*, and may create congestion for the network
- Ex: The post office does its best to deliver the regular mail but does not always succeed. If an unregistered letter is lost or damaged, it is up to the sender or would-be recipient to discover this.
- Opposite of best effort is dedicated service
- **IP relies on upper layer transport protocols (TCP) to take care of these problems. Hence we write TCP/IP!**

Encapsulation/Decapsulation



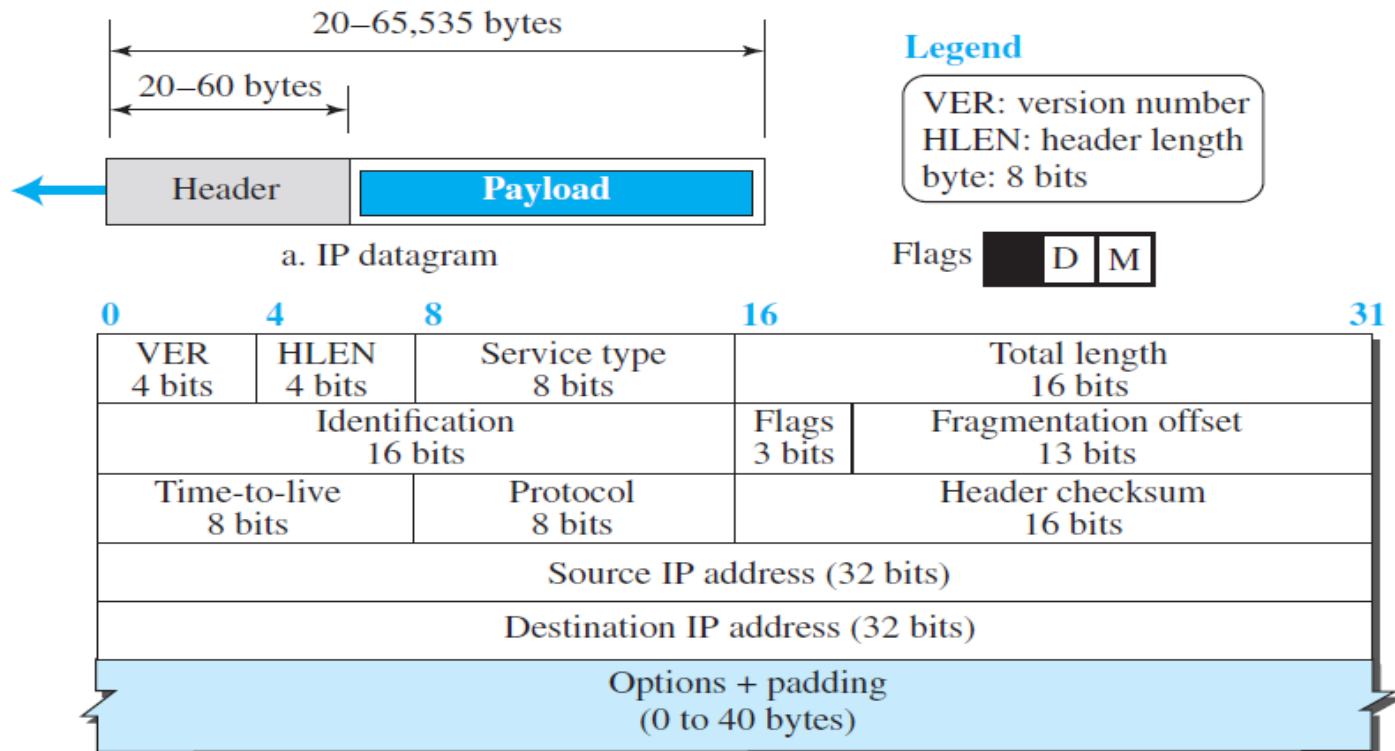
Datagram Format

Header size: 20 (min)-60 (max)bytes

Payload (from higher layer): 0-65515 bytes

$20 + 65515 = 65535$

Figure 19.2 *IP datagram*



Datagram Format

- **VER:** the version of the IPv4 protocol, which, obviously, has the value of 4 (0100)
- **HLEN:** defines the total length of the datagram header (4 bits)

0000 (0)

0001 (1)

0101 (5) 5*4=20 bytes



1111(15) 15*4= 60 bytes

- **Service Type:** which defined how the datagram should be handled (8 bits).
Field to provide *differentiated services*. (8 bits)
 - Used to identify which traffic should receive priority treatment
 - Precedence(3) (to set priority- whether the packet is routine(normal) or immediate)
 - Type of Service(5) (Delay , Throughput, Reliability, Cost, Reserved)

Example 19.1

Q. An IPv4 packet has arrived with the first 8 bits as shown: 01000010. The receiver discards the packet. Why?

- There is an error in this packet. The 4 leftmost bits (0100) show the version, which is correct.
- The next 4 bits (0010) show an invalid header length ($2 \times 4 = 8$). The minimum number of bytes in the header must be 20. The packet has been corrupted in transmission.

Datagram Format

- **Total Length:** The total length of the datagram including the header.
 - A 16-bit number can define a total length of up to 65,535
 - $\text{Length of data} = \text{total length} - (\text{HLEN}) \times 4$
 - $65535 - (5 \times 4) = 65515$
- **Identification, Flags, and Fragmentation Offset**

These three fields are related to the fragmentation of the IP datagram when the size of the datagram is larger than the underlying network can carry.
- **Time-to-live (8 bits i.e. 0 to 255)**
- (TTL) field is used to control the maximum number of hops (routers) visited by the datagram. When a source host sends the datagram, it stores a number in this field.
- Each router that processes the datagram decrements this number by one. If this value, after being decremented, is zero, the router discards the datagram.

Example 19.3

In an IPv4 packet, the value of HLEN is 5, and the value of the total length field is $(0028)_{16}$. How many bytes of data are being carried by this packet?

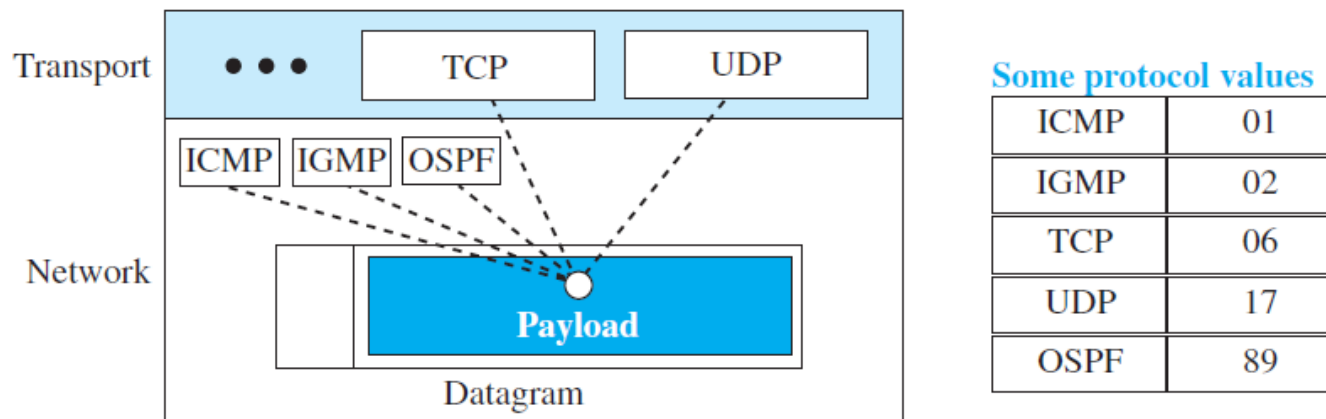
- The HLEN value is 5, which means the total number of bytes in the header is 5×4 , or 20 bytes (no options).
- The total length is $(0028)_{16}$ or 40 bytes,

$$\begin{aligned}\text{Length of data} &= \text{total length} - (\text{HLEN}) \times 4 \\ &= 40 - 20 = 20 \text{ bytes}\end{aligned}$$

Datagram Format

- **Protocol:** the data section of a packet, called the *payload*, carries the whole packet from another protocol.
- The Internet authority has given any protocol that uses the service of IP a unique 8-bit number which is inserted in the protocol field.
- When the payload is encapsulated in a datagram at the source IP, the corresponding protocol number is inserted in this field; when the datagram arrives at the destination, the value of this field helps to define to which protocol the payload should be delivered.

Figure 19.3 *Multiplexing and demultiplexing using the value of the protocol field*



Datagram Format

Header checksum: IP adds a header checksum field to check the header, but not the payload

- IP is not a reliable protocol; it does not check whether the payload carried by a datagram is corrupted during the transmission
- Errors in the IP header can be a disaster.
 - For example, if the destination IP address is corrupted, the packet can be delivered to the wrong host.
 - If the protocol field is corrupted, the payload may be delivered to the wrong protocol.
 - If the fields related to the fragmentation are corrupted.

Datagram Format

Source and Destination Addresses. (32-bit)

- Define the IP address of the source and destination respectively
- The source host should know its IP address. The destination IP address is either known by the protocol that uses the service of IP or is provided by the DNS
- The value of these fields must remain unchanged during the time the IP datagram travels from the source host to the destination host.

Options

- 40 bytes of options.
- Network testing and debugging

Payload

- Payload, or data, is the main reason for creating a datagram.
- Payload is the packet coming from other protocols that use the service of IP

Example 19.2

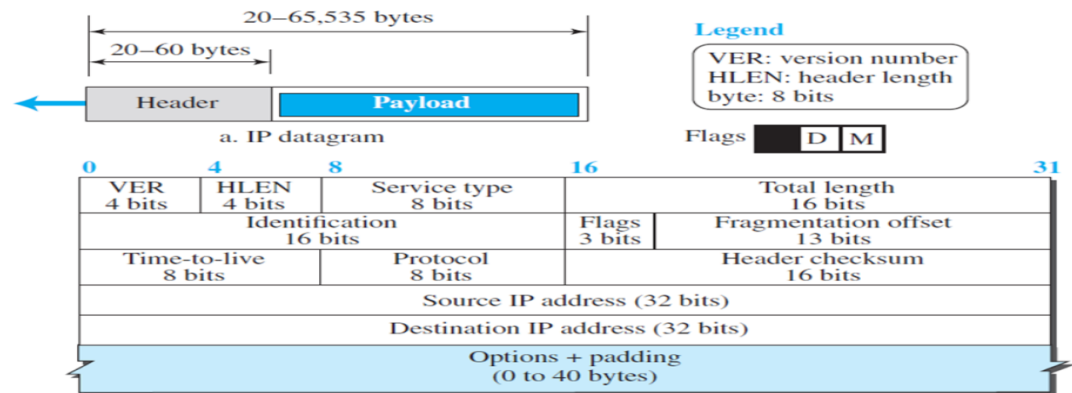
In an IPv4 packet, the value of HLEN is $(1000)_2$. How many bytes of options are being carried by this packet?

- The HLEN value is 8, which means the total number of bytes in the header is 8×4 , or 32 bytes.
- The first 20 bytes are the base header, the next 12 bytes are the options.

$$20 \text{ (base header)} + 12 \text{ (options)} = 32$$

Figure 19.2 IP datagram

Example 19.4



An IPv4 packet has arrived with the first few hexadecimal digits as shown.
 (45000028000100000101...)₁₆

How many hops can this packet travel before being dropped? The data belong to what upper-layer protocol?

- To find the time-to-live field, we skip 8 bytes (16 hexadecimal digits).
 $8 \times 8 = 64 \text{ bits} / 4 \text{ bits} = 16$
- The time-to-live field is the ninth byte, which is (01)₁₆. This means the packet can travel only one hop.
- The protocol field is the next byte (02)₁₆, which means that the upper-layer protocol is IGMP.

Example 19.5

Figure 19.4 shows an example of a checksum calculation for an IPv4 header without options. The header is divided into 16-bit sections. All the sections are added and the sum is complemented after wrapping the leftmost digit. The result is inserted in the checksum field.

Figure 19.4 *Example of checksum calculation in IPv4*

4	5	0	28	
49.153			0	0
4	17	0		
10.12.14.5				
12.6.7.9				

4, 5, and 0

28

1

0 and 0

4 and 17

0

10.12

14.5

12.6

7.9

Sum

Wrapped sum

Checksum

→

→

→

→

→

→

→

→

→

→

→

→

→

4

5

0

0

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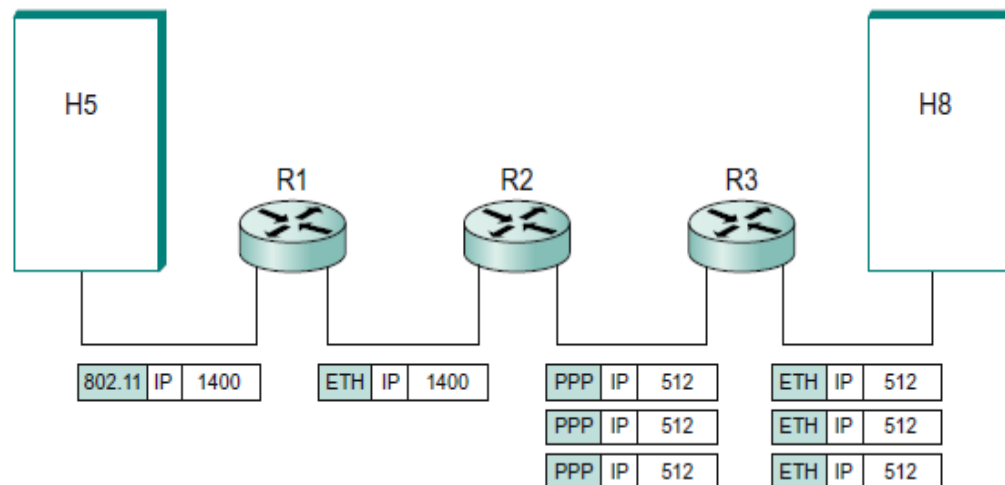
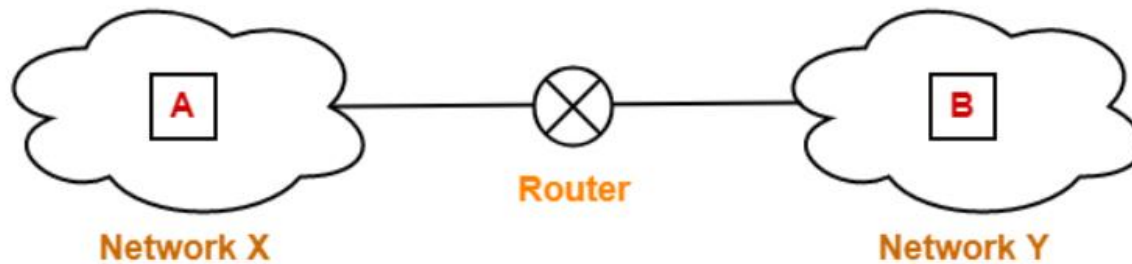
Note that the calculation of wrapped sum and checksum can also be done as follows in hexadecimal:

Wrapped Sum = Sum mod FFFF

Checksum = FFFF – Wrapped Sum

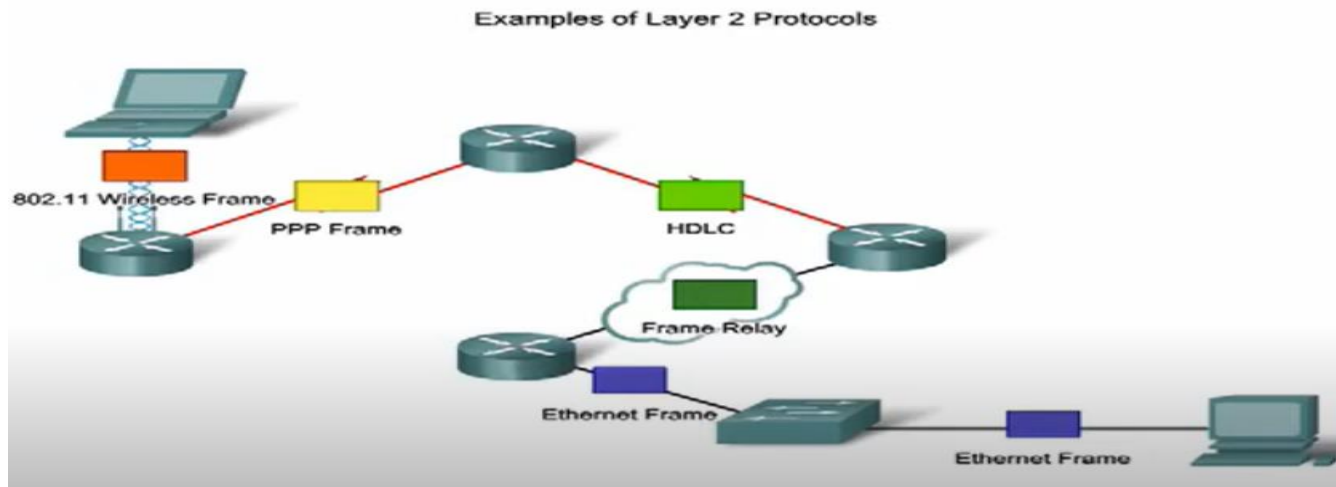
Fragmentation

- IP Fragmentation is a process of dividing the datagram into fragments during its transmission.
- It is done by intermediary devices such as routers at the destination host at network layer.



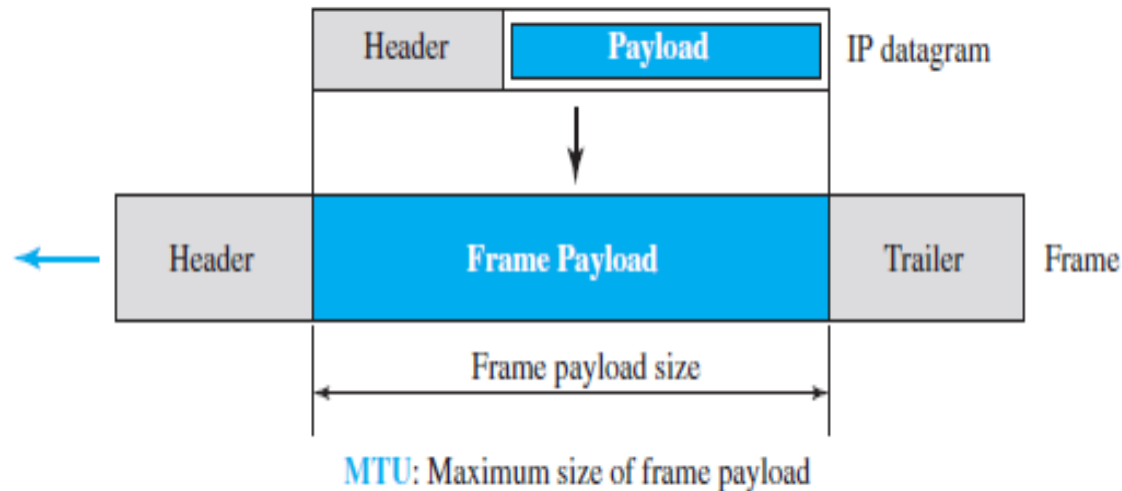
Fragmentation

- A datagram can travel through different networks.
- Each router decapsulates the IP datagram from the frame it receives, processes it, and then encapsulates it in another frame.
- Format and size of the received frame depend on the protocol used by the physical network through which the frame has just traveled.
- For Example:
 - If a router connects a LAN to a WAN, it receives a frame in the LAN format and sends a frame in the WAN format.



Maximum Transfer Unit (MTU)

- Each Link-layer protocol has its own frame format.
- Format is the maximum size of the payload that can be encapsulated.
 - Datagram is encapsulated in a frame , should be maximum size.



MTU differs from one physical network protocol to another.

Ex: LAN, Ethernet is normally 1500 bytes, but for a WAN it can be larger or smaller

Maximum length of the IP datagram is 65,535 bytes.

Fragmentation

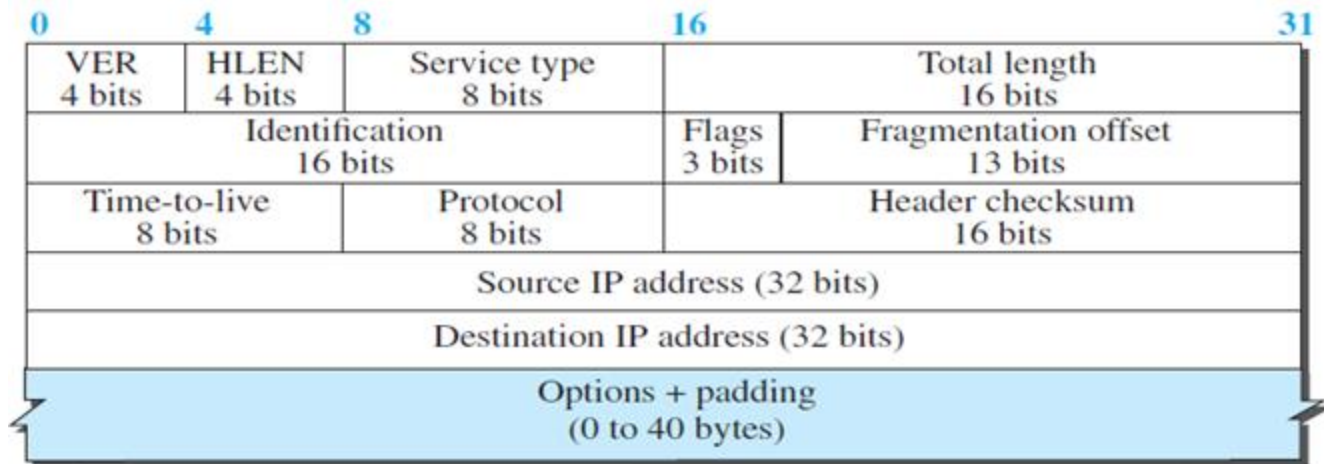
- We must divide the datagram to make it possible for it to pass through these networks is called fragmentation
- This fragmentation process takes place at the IP layer (OSI layer 3) and marks the packets it fragments as such, so that the IP layer of the destination host knows it should reassemble the packets into the original datagram.

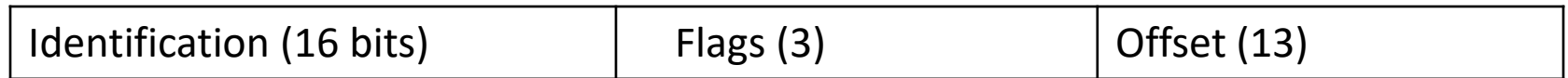
Reassembly

- A datagram can be fragmented by the source host or any router in the path.
- The *reassembly of the datagram, however, is done only by the destination host because* each fragment becomes an independent datagram.
- Fragmented datagram can travel through different routes, and we can never control or guarantee which route a fragmented datagram may take, all of the fragments belonging to the same datagram should finally arrive at the destination host

Fields Related to Fragmentation

- Three fields in an IP datagram are related to fragmentation:
 - *Identification.*
 - *Flags.*
 - *Fragmentation offset.*





Divided one packet into 3 datagrams
(which may take different routes)

To identify all the three datagrams belongs
to one packet, identification field is used.

2^{16} possible values

0 – There is a fragment
1 – There is no fragment

1 or 0
It is a first packet
or last packet

Number of bytes ahead of it

Identification

- The 16-bit identification field identifies a datagram originating from the source host.
- The combination of the identification and source IP address must uniquely define a datagram as it leaves the source host.
- When a datagram is fragmented, the value in the identification field is copied into all fragments

Flags

- **3-bit *flags field* defines three flags.**
- The leftmost bit is reserved (not used).
- The second bit (D bit) is called the *do not fragment bit*.
 - If its value is 1, the machine must not fragment the datagram.
 - If its value is 0, the datagram can be fragmented if necessary.
- The third bit (M bit) is called the more fragment bit
 - If its value is 1, it means the datagram is not the last fragment; there are more fragments after this one. If its value is 0, it means this is the last or only fragment.

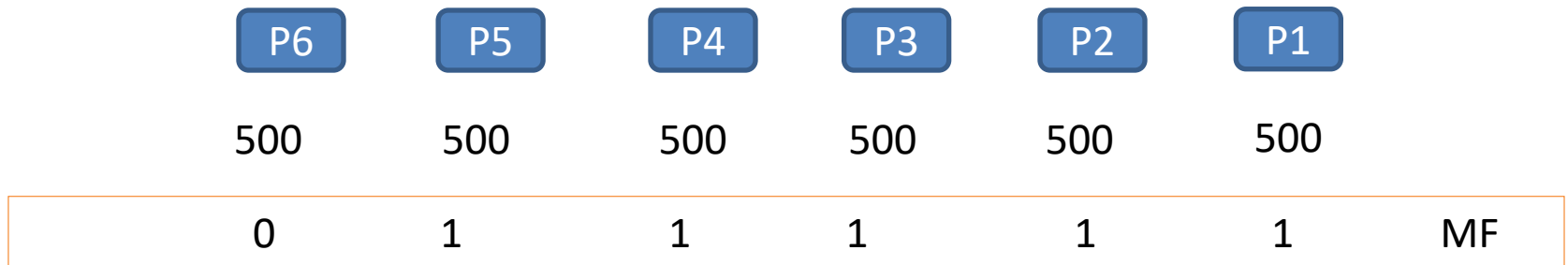
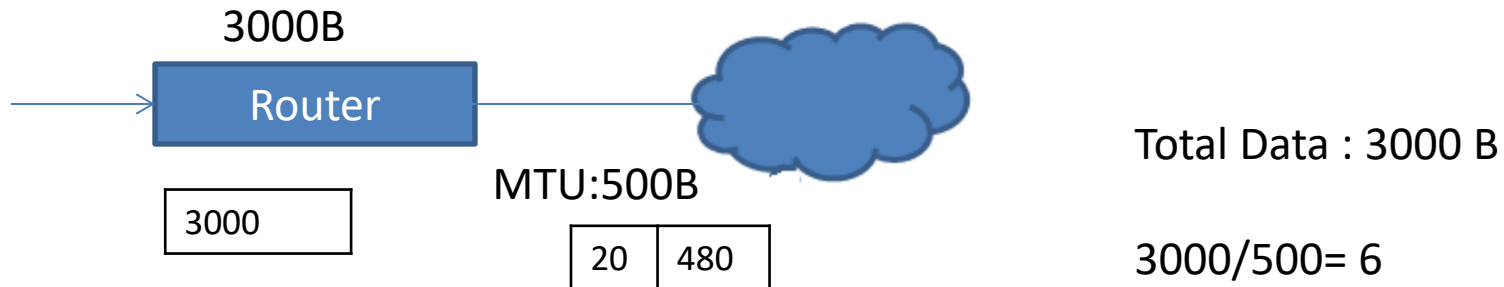
offset field

- **13-bit fragmentation offset field.**
- Shows the relative position of this fragment with respect to the whole datagram.
- Number of bytes ahead of it

P3 200 bytes	P2 200 bytes	P1 200 bytes
-----------------	-----------------	-----------------

- First part of the original datagram - It sets the Offset to $0/8=0$
- The second fragment – **$200/8=25$**
- **Third fragment – $200+200/8=50$**
 - (*Scaling factor: 8.* It is the offset of the data in the original datagram measured in units of 8 bytes)

A datagram of 3000 B reached at router and must be forward to link with MTU of 500 B. How many fragments will be generated and also write MF, offset, total length value for all



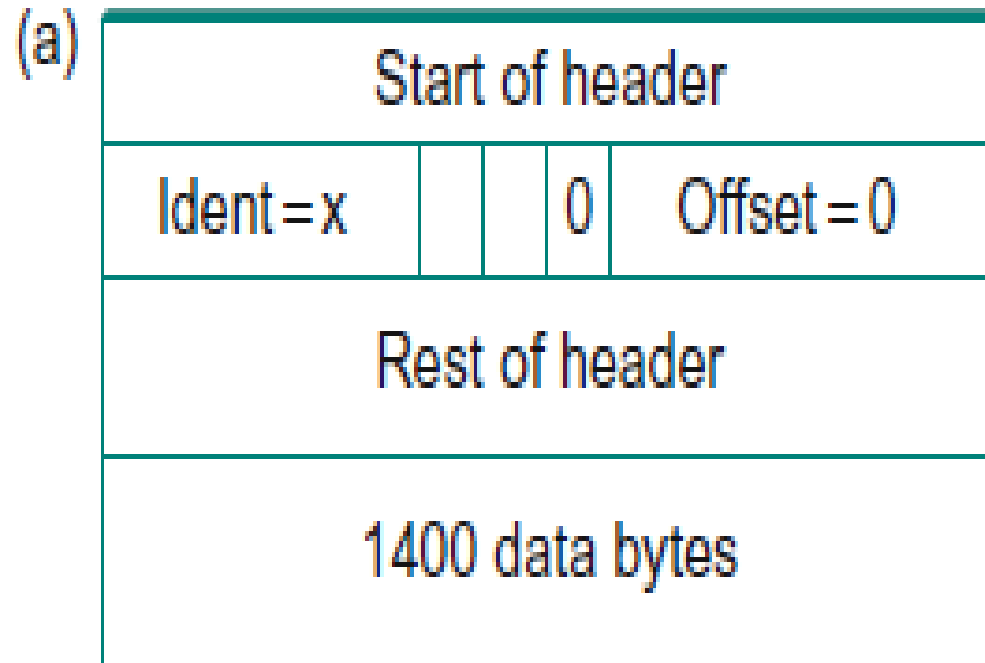
(Last packet is not followed by any other packet)

$$500/8=62.5 \quad 0/8=0$$

Offset

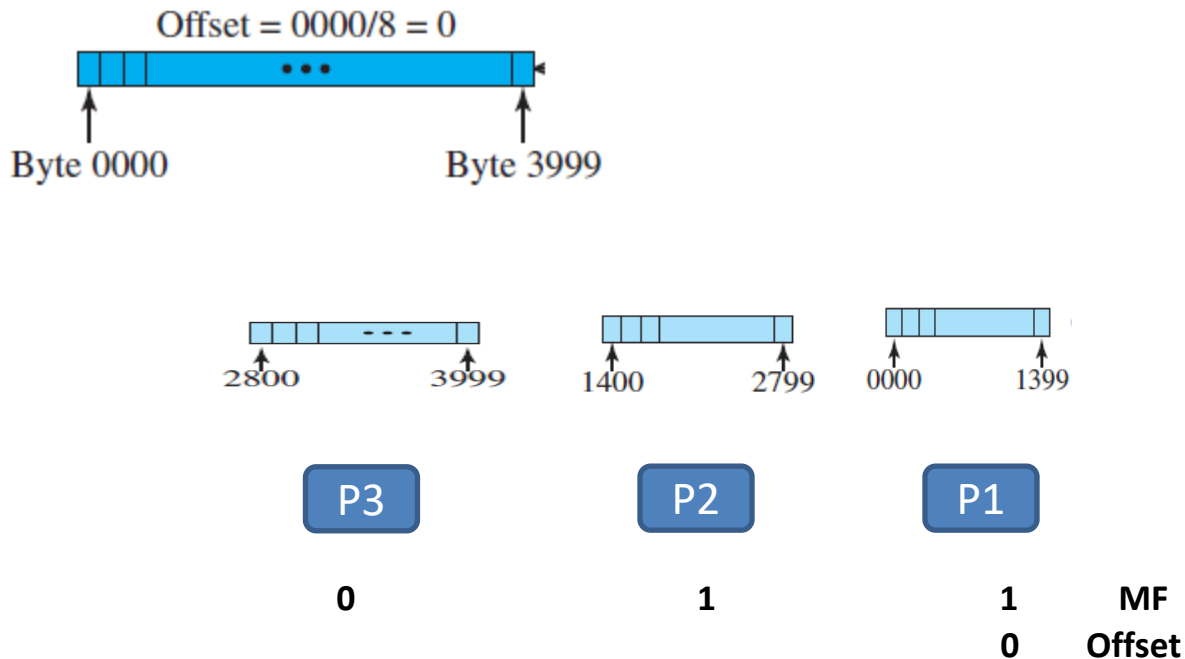
Number of bytes ahead of it

Unfragmented packet



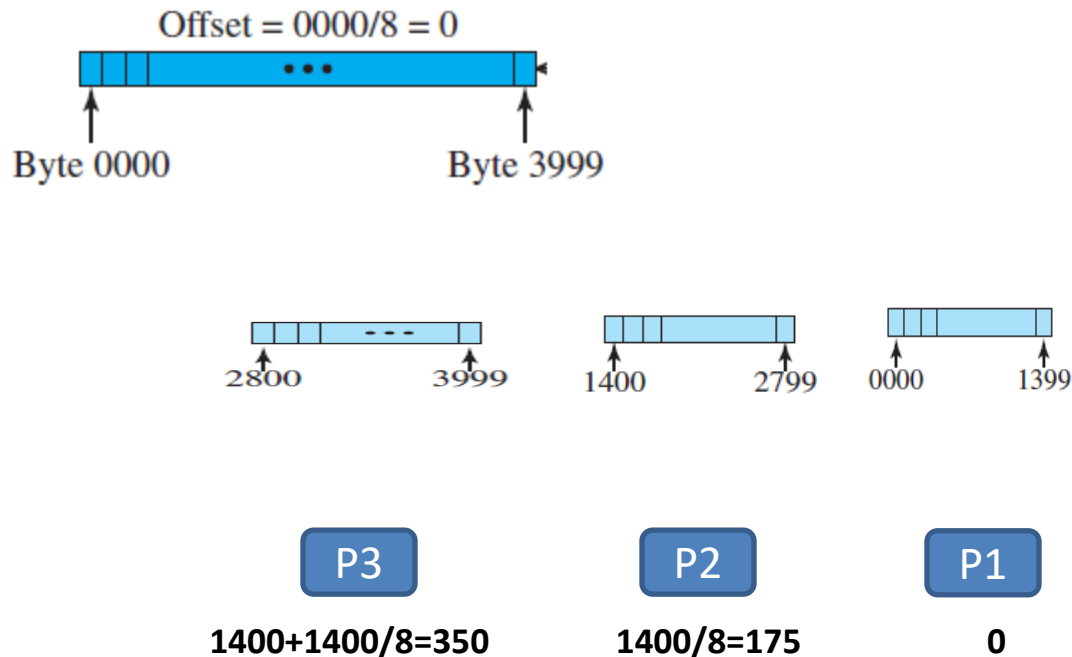
Fragmentation Example

- Shows a datagram with a data size of 4000 bytes.
- Fragmented into three fragments.
- Original datagram are numbered 0 to 3999.



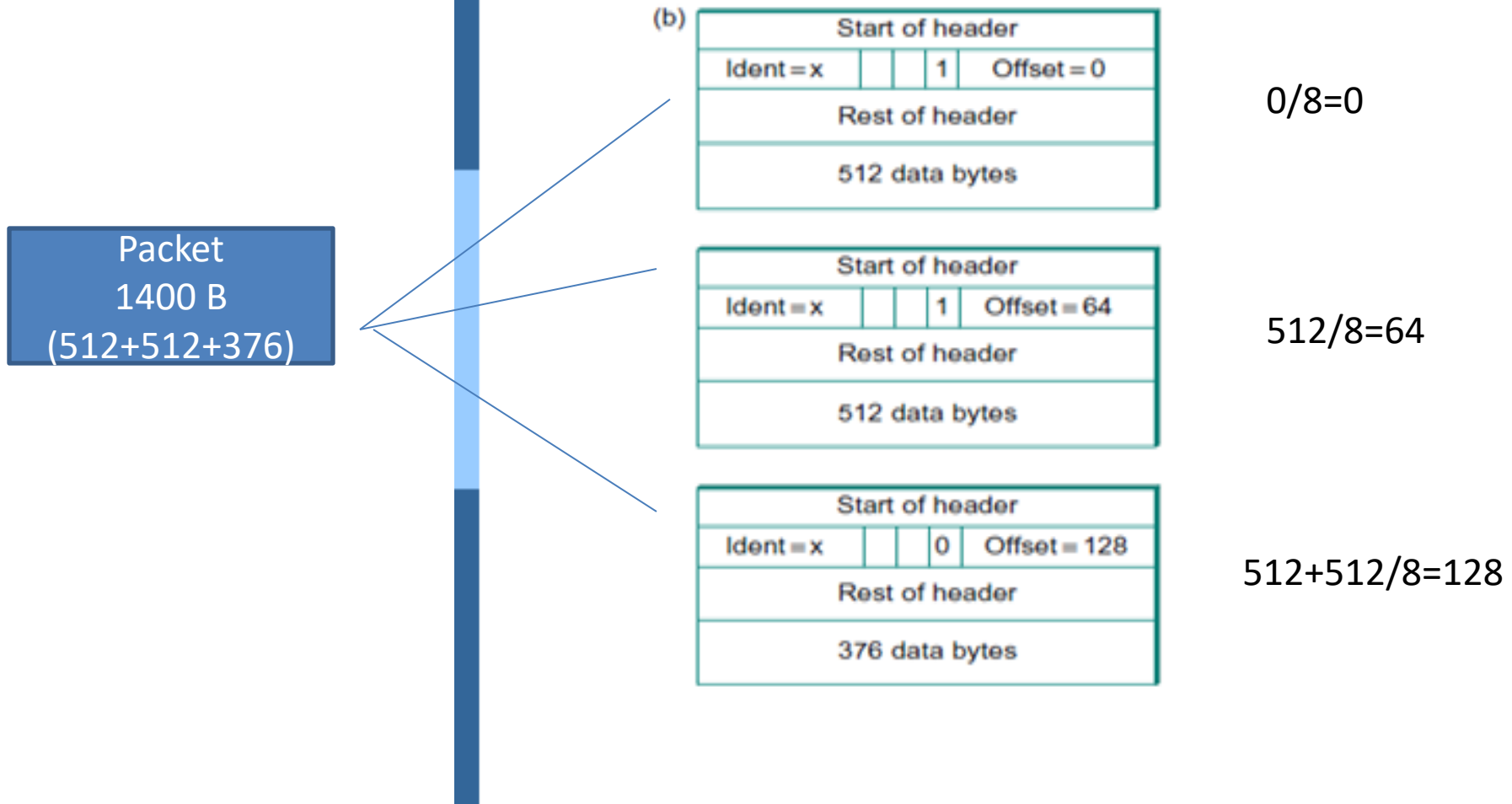
Fragmentation Example

- Shows a datagram with a data size of 4000 bytes.
- Fragmented into three fragments.
- Original datagram are numbered 0 to 3999.

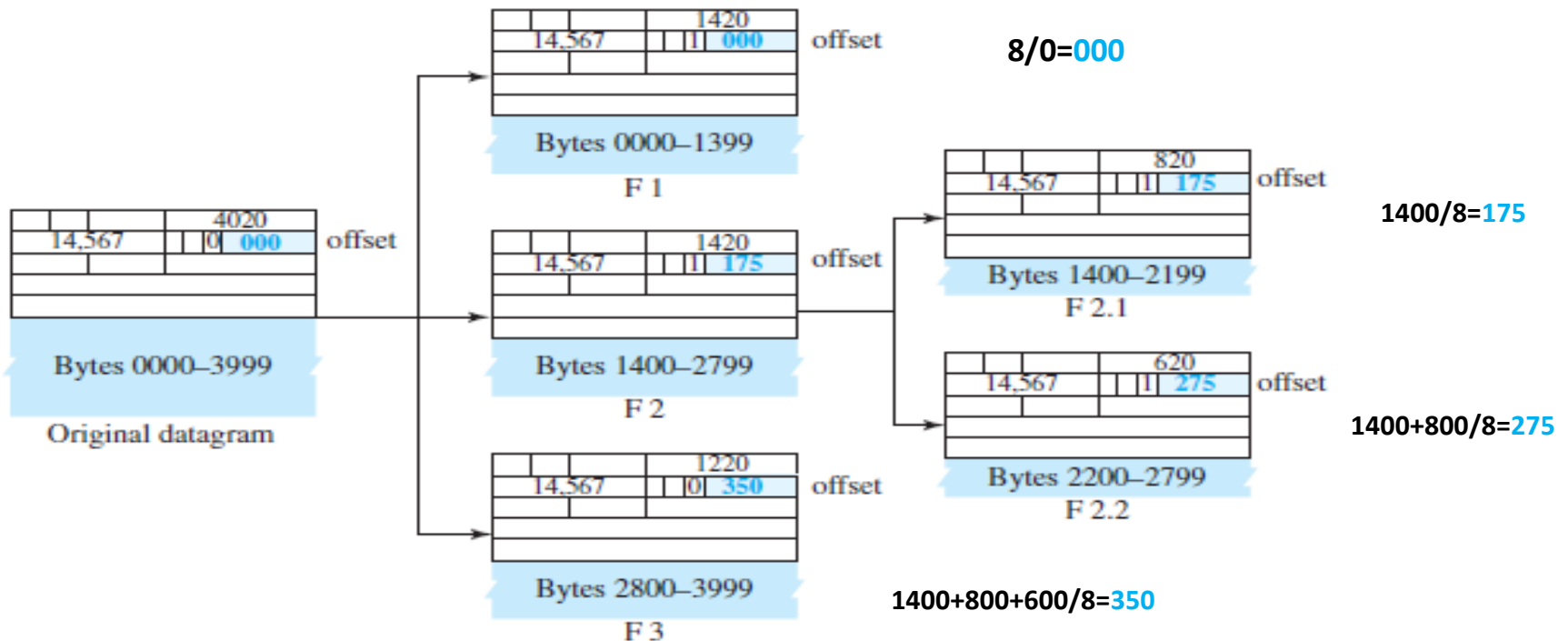
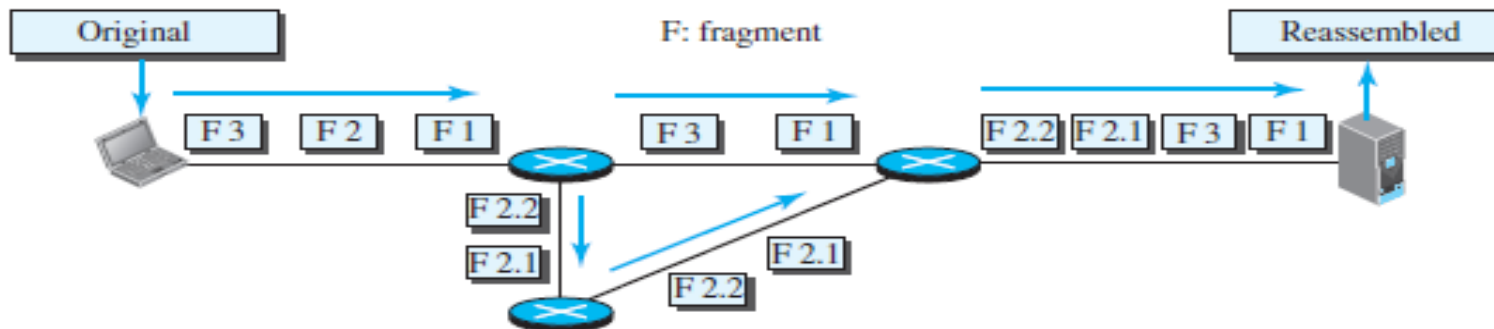


Fragmentation Example

Fragmented packets



Detailed fragmentation example



Example 19.6

A packet has arrived with an M bit value of 0. Is this the first fragment, the last fragment, or a middle fragment? Do we know if the packet was fragmented?

Solution

- If the M bit is 0, it means that there are no more fragments; the fragment is the last one.
- However, we cannot say if the original packet was fragmented or not. A nonfragmented packet is considered the last fragment.

Example 19.7

A packet has arrived with an M bit value of 1. Is this the first fragment, the last fragment, or a middle fragment? Do we know if the packet was fragmented?

- If the M bit is **1**, it means that there is at least **one more fragment**. This fragment can be the first one or a middle one, but not the last one.
- We don't know if it is the first one or a middle one; we need more information (the value of the fragmentation offset).

Example 19.8

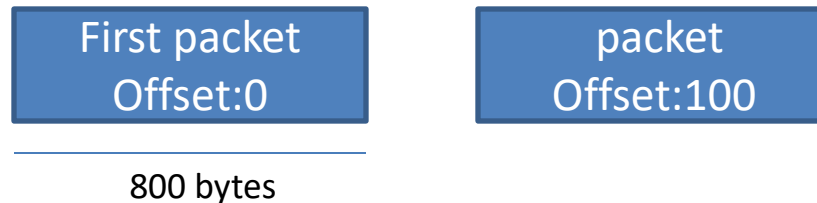
A packet has arrived with an M bit value of 1 and a fragmentation offset value of 0. Is this the first fragment, the last fragment, or a middle fragment?

Solution

- Because the M bit is 1, it is either the first fragment or a middle one. Because the offset value is 0, it is the first fragment.

Example 19.9

A packet has arrived in which the offset value is 100. What is the number of the first byte? Do we know the number of the last byte?



Solution

- To find the number of the first byte, we multiply the offset value by 8. This means that the first byte number is 800.
- We cannot determine the number of the last byte unless we know the length of the data.

Example 19.10

A packet has arrived in which the offset value is 100, the value of HLEN is 10, and the value of the total length field is 100. What are the numbers of the first byte and the last byte?



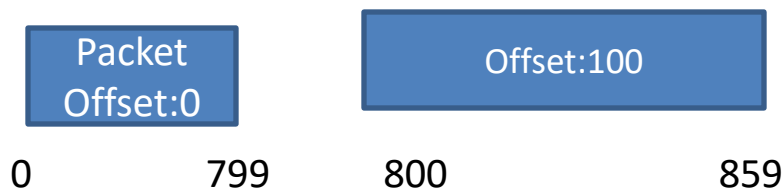
Solution

800 bytes

- The first byte number is $100 \times 8 = 800$.
- The total length is 100 bytes, and the header length is 40 bytes (10×4), which means that there are 60 bytes in this datagram.

Hlen ($10 \times 4 = 40$) Total length: $100 - 40 = 60$

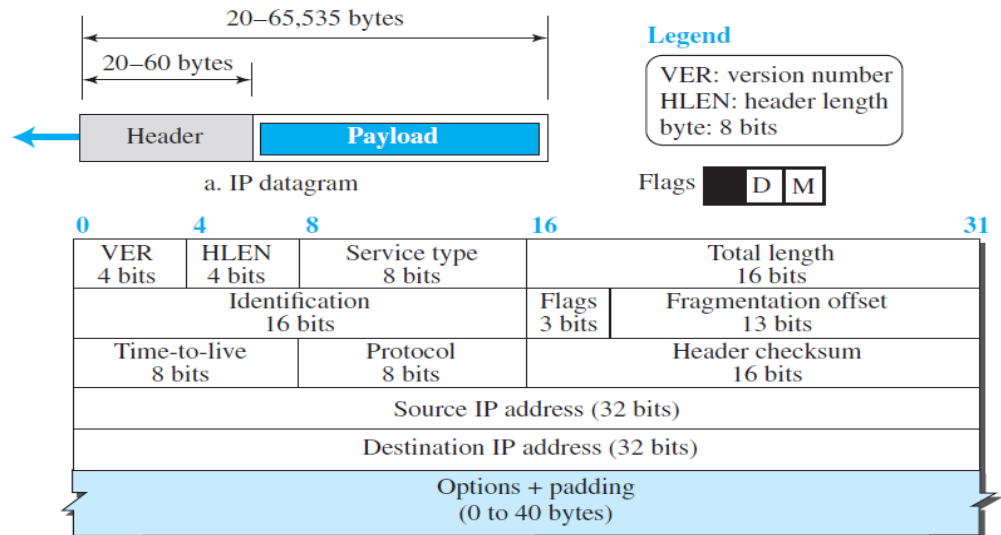
- If the first byte number is 800, the last byte number must be 859.



Options

- IPv4 datagram is made of two parts:
 - Fixed part (20 bytes)
 - Variable part
- Variable part comprises the options that can be a maximum of 40 bytes.
- Options, as the name implies, are not required for a datagram.
- Network testing and debugging.

Figure 19.2 IP datagram



Options

- Options are divided into two broad categories:
 - Single-byte options
 - Multiple-byte options
- Single-byte options: No Operation and End of options.
- Multiple-byte options: There are four multiple-byte options.
 - Record Route
 - Strict Source Route
 - Loose Source Route
 - Timestamp
- *A record route option is used to record the Internet routers that handle the datagram.*
- *A strict source route option is used by the source to predetermine a route for the datagram as it travels through the Internet.*
- *A loose source route option is similar to the strict source route. Each router in the list must be visited, but the datagram can visit other routers as well.*
- *A timestamp option is used to record the time of datagram processing by a router.*

Security of IPv4 Datagrams

- There are three security issues that are particularly applicable to the IP protocol:
 - Packet sniffing
 - Packet modification
 - IP spoofing.

Packet Sniffing

- An intruder may intercept an IP packet and make a copy of it.
- Packet sniffing is a passive attack.
- Packet sniffing cannot be stopped.
- Encryption of the packet can make the attacker's effort useless.
- May still sniff the packet, but the content is not detectable.

Packet Modification

- The attacker intercepts the packet, changes its contents, and sends the new packet to the receiver.
- Type of attack can be detected using a data integrity mechanism.

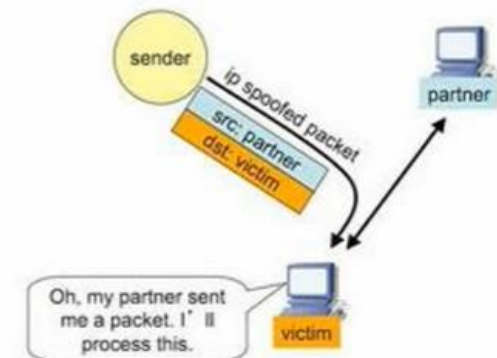
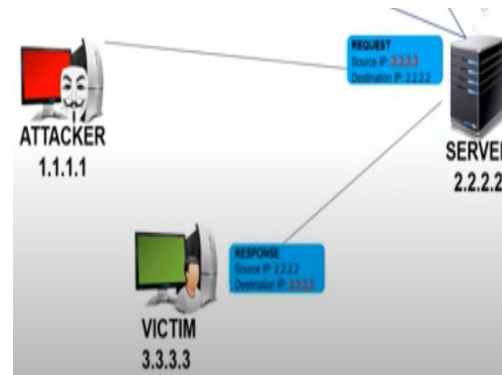
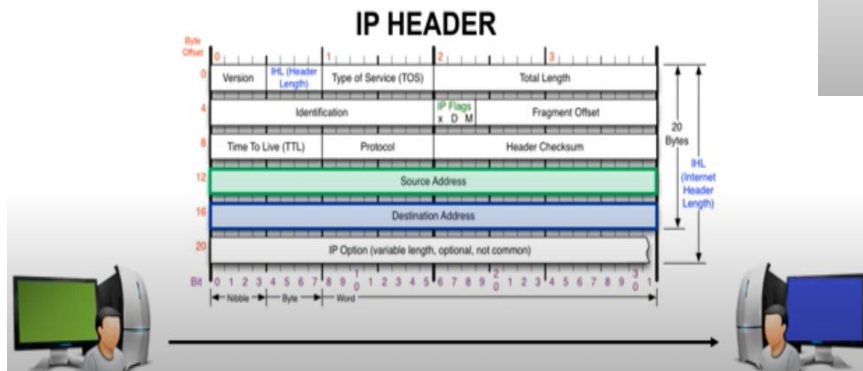
IP Spoofing

- **IP Spoofing** is a technique used to gain unauthorized access to machines.
 - Ex. An attacker can send an IP packet to a bank pretending that it is coming from one of the customers

IP Spoofing involves modifying the packet header with a forged (**spoofed**) source **IP** address, a checksum, and the order value.

Aiming to transfer packets between two ends, the IP header contains:

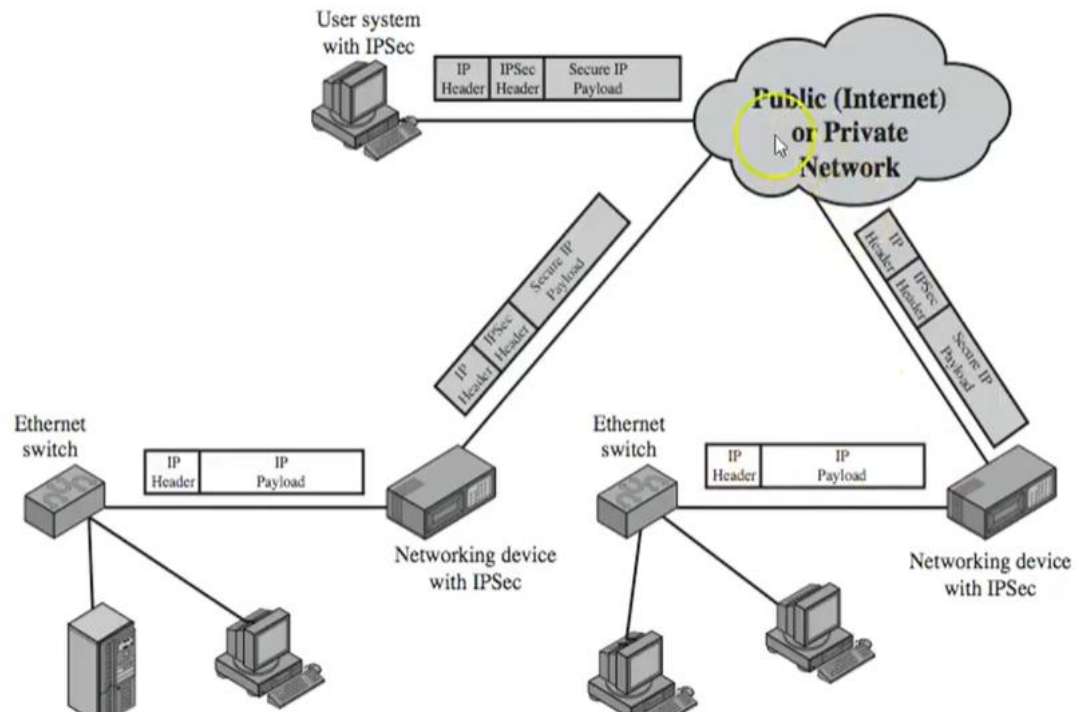
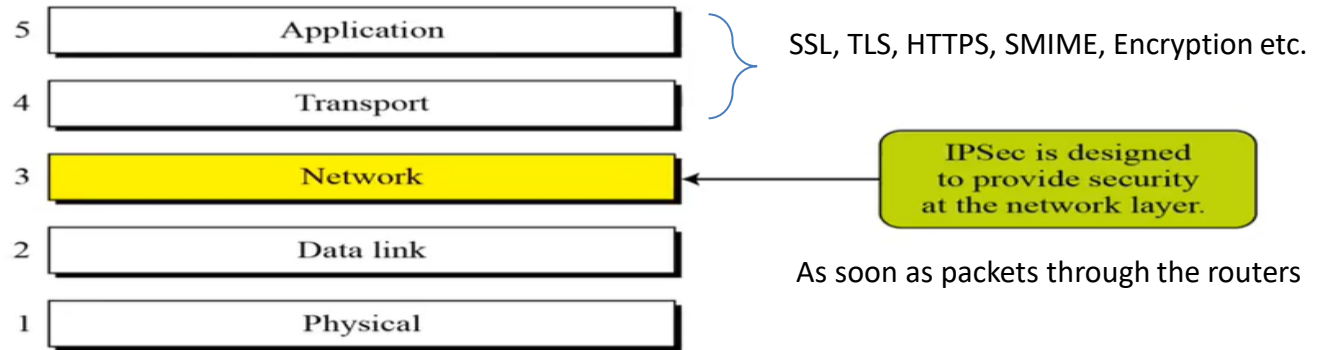
- **Source Address** – indicating **from** where the packet was sent.
- **Destination Address** – indicating **to** where the packet was sent.



IPSec

- IP packets today can be protected using a protocol called IPSec (IP Security).
- This protocol, which is used in conjunction with the IP protocol, which provides secure exchange of packets at the IP layer.
- IPSec provides the following four services:
 - Defining Algorithms and Keys: The two entities that want to create a secure channel
 - Packet Encryption: The packets exchanged between two parties can be encrypted
 - Data Integrity: Data integrity guarantees that the packet is not modified during the
 - Origin Authentication : IPSec can authenticate the origin of the packet

Example of IPSec



References

- Forouzan, A. Behrouz. *Data communications & networking (sie)*. Tata McGraw-Hill Education, 5E